Indira Gandhi National Open University
School of Management Studies

MMPM – 003
Product and Brand Management

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Acknowledgement: Relevant parts of this course have been adapted and updated from the course MS 63: Product Management. MS 63 course was prepared by the experts (names mentioned above in Italics) and their profile is as it was in that material.

MATERIAL PRODUCTION

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Assistant Registrar
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July, 2022

ISBN:

Printed and published on behalf of the Indira Gandhi National Open University, New Delhi, by the Registrar, MPDD, IGNOU.

Printed at : M/s Educational Stores, S-5 Bulandshahar Road Industrial Area, Site-1, Ghaziabad (UP)-201009

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Block

1

INTRODUCTION TO PRODUCT MANAGEMENT

BLOCK 1 INTRODUCTION TO PRODUCT MANAGEMENT

Products form a tangible expression of all that an enterprise has to offer. It is one's experience with the products of an enterprise that largely influences a consumer evaluation of business endeavour and affects his future patronage. Product management therefore becomes central and core to the marketing management function in the firm.

Block 1 form the introductory block of this course on product and brand management and consists of 4 units.

The first unit familiarizes you with the term product and its related concepts and the product planning activities in an organization.

In the second unit you will be introduced to the concept of Product Life Cycle (PLC) its characteristics, objectives and strategies which helps you understand the role of PLC in product management decision making.

The third unit gives you a fairly good idea to understand and appreciate the concept and logic behind product lines, the reasons for the proliferation of product lines and their merits and limitations.

The last unit focuses on the strategic importance of managing products continuously with a vision, in response to the ever changing needs and wants of the end users and prevailing market forces. Precisely, this unit focuses on the firm's product portfolio its models for decision making.

UNIT 1 PRODUCT MANAGEMENT BASIC CONCEPTS

Objectives

After reading this unit you should be able to:

- discuss on the meaning and scope of product management
- define a product offering
- explain the types of products ,their nature, classification, differentiation and product
- and brand relationships.
- appreciate the origin and scope of product management and product planning tasks
- understand the role and responsibilities of a product manager
- comprehend the conceptual issues.

Structure

1.1 INTRODUCTION

Product is core to any business enterprise and forms the most tangible expression offered by the firm in response to consumer needs and wants. Issues pertaining to `product', therefore, become key determinants of successful business strategy.

In this unit you will be introduced to the meaning, scope and major areas of responsibilities of the product management function within the organisation. The unit also familiarizes you with what a product is all about, mentioning the related conceptual issues related to Product / Product Management.

1.2 PRODUCT MANAGEMENT – ITS MEANING AND SCOPE

Among the 4 Ps of Marketing mix elements which you are familiar with the first P i.e. ***"Product"*** is the main carrier of consumer value. Product constitutes the core of the offer made by the firm. It may be a car, an air ticket or an app. You cannot imagine any role or relevance of the other three elements of the marketing mix i.e. price, place and promotion in isolation from the product. They are also important carriers of consumer value and their complementarity and compatibility with the product is essential in designing the firm's business strategy and making the product a success in the market place.

How a firm conceives and creates this product and successfully markets it to the customer is the theme of Product Management. Product Management is the holistic job of Product Managers which includes planning, forecasting and marketing of products and services

Every professionally managed progressive and pro-active manufacturing and marketing firm which responds to market needs resorts to a Product Planning exercise in relation to its customers, relevant markets, competition and other prevailing market forces.

The task of Product Planning derives from the Marketing Plan which is a functional sub-set of the overall Corporate Plan. In product planning are included product strategies relating to product line -length and breadth and line stretching, both upwards and downwards.

The scope of Product Management also considers conceptual issues like the Product Life Cycle, New Product Development related issues to Pricing, Promotion and Distribution and the Product Portfolio.

1.3 PRODUCT MANAGEMENT – AN OVERVIEW

In simple terms, Product Management means managing various product lines/ products and the overall product mix of the company. Product Management is also known by the term Product Policy.

The major/main tasks in Product Management include the following:

- Appraisal of each product line and each product/brand in the line
- Decisions on packaging
- Product differentiation and positioning
- Managing brands and developing brand equity
- Innovations and new product development
- Managing product quality
- Managing the Product Life Cycle of products/ brands

1.4 WHAT IS A PRODUCT?

The term "Product" in marketing means any offer that a company brings out for sale. It could be beauty soap, a laptop, a holiday package, or even a wallet service by an e- payment provider. "Product" is a common term to denote the company's offer to the market. It is by offering something to the consumer that the firm begins its marketing task. The job of marketing is to make the product and the customer meet and result in a continuing relationship.

A Product may be a physical good, service or an idea, or any combination of the three, offering a bundle of satisfactions. Pepsi selling Lay's chips is selling a product, advertising agency JWT provides services to organizations by preparing advertising campaigns ,and Vandana Luthra sells ideas to lose weight as well as skin -care products.

Taking the idea further, the physical entity is only one component of the product personality. The various features and functions built around it – the brand name, the package and labelling, the quality associated with it, the guarantees, the price label, the manufacturer's name and prestige-all go into the personality of the product.

According to *Theodore Levitt*-"Products are almost and always a combination of the tangible and the intangible. For example an automobile is not simply a machine for movement visibly or measurably differentiated by design, size, colour, options, horse-power, or kilometre per litre. It is also a complex symbol denoting status, taste, rank, achievement, aspiration, and (these days) being "smart"—that is, buying fuel economy rather than display. But the customer buys even more than these attributes.

Besides, the enormous efforts of the auto manufacturers to cut the time between placement and delivery of an order and to select, train, supervise, and motivate their dealerships suggest that these too are integral parts of the products people buy and are, therefore, ways by which products may be differentiated to the potential buyer.

Thus a product is a complex cluster of value satisfactions. The generic thing is not itself the product; it is merely, as in poker, table stakes—the minimum that is necessary at the outset to give its producer a chance to play the game. It is the playing that gets the results, and, in business, this means getting and keeping customers. Customers attach value to a product in proportion to its perceived ability to help solve their problems or meet their needs. In the words of a specialist in industrial marketing has expressed it, "The "'product" is the total package of benefits the customer receives when he or she buys."

When Apple sells an I Phone, or Chanel sells a perfume, are they selling a rectangular box or bottled perfumed water? Is that for what consumers pay fancy prices willingly? The question is clearly rhetorical, and the obvious answer is an emphatic no. Instead, what these companies are really selling is the satisfaction, use or benefit the consumer wants.

A product essentially means the need-satisfying offering of a firm. The idea of "Product' as potential customer satisfaction or benefits is very important. Many business managers get wrapped up in the technical details involved in producing a product. But that is not how customers view the product. Most customers think about product in terms of the total satisfaction it provides. That satisfaction may require a complete product offering that is really a combination of excellent service, a physical good with the right features, useful instructions, a convenient package, a trustworthy warranty or familiar name that has positive memories for the customer.

According to Philp Kotler "At the heart of great brand is a great product. To achieve market leadership, firms must offer products and services of superior quality that provide unsurpassed customer value"

1.5 PRODUCT CHARACTERISTICS AND CLASSIFICATION

Various components of a Product can be conveniently classified as follows:

- The Core product
- The Associated features
- The Brand name
- The logo
- The package and label

The Core Product

The core or basic constituent of the product is the first component. By taking the example of Mysore Sandal soap. The fragrance of sandal oil, the oval shape, the sandal colour, the brand name, the price, the place of manufacture, the positioning as luxury soap everything has gone into the making of the product personality. But the core component is the soap, the generic constituent, as in the case of any other bath soap the other components are superimposed on this basic constituent to develop the total personality of Mysore Sandal Soap. The first lesson in Product Management is that the basic product constituent has to be good for the total product personality to be effective.

The Associated features

To continue with the above example of Mysore Sandal Soap the associated features are its fragrance, the oval shape, the sandal colour and the like which contribute to the total personality of the product. Most attempts of marketers to enhance the product personality are centred on the associated features. Even when two products share the same plank in respect of the core constituent and function, differences in associated features are the basis for differentiation. While in the case of consumer durables, product design is the big value addition from smart phones to toys. Apple has built up its entire business on its unique competence at design.

Product features vs. Product benefits

Product features are only the beginning; it is the benefits that they yield which are of relevance to the consumer.

Brand name

We live in the age of brands. In fact, the brand name is a major selling tool. The intensive brand promotion exercises conducted by brands have made consumers extremely brand conscious. Today, we do not ask for toothpaste or a cell phone or a cupboard or a perfume - we ask for a Colgate or a One Plus or Godrej or a Christian Dior. People trust brands and pay disproportionate price (in relation to the cost of manufacturing the product) for acquiring the brand of their choice. It would not be out of place to say that the marketer adorns or beautify the product with the brand.

The Logo

The logo or the brand mark /symbol, too is an integral part of the product. It supports the brand effectively. Symbols and pictures serve the purpose of product / brand identification and recall. The logo's importance is enhanced in rural markets where less learned customers may form a large chunk and the written matter on the packaging is of no use to them so they recognize the product by the picture in the logo. Brands such as Nirma and Dettol (below) are recognized in rural areas by the visuals incorporating the logos.

The Nirma girl

The Package

The Package is so important that some marketers have even given it the status of the **5th P** or fifth element of marketing mix. The package serves three essential roles:

1. It protects the product
2. It provides information about the product
3. It adds to its aesthetics and sales appeal.

From being merely a protective cover, packaging has evolved into a major game changer and a great value- addition. The material of the package, the colour, its shape and size, its finish, its labelling, the possibilities of its re-use often come together to enhance the sales appeal of the product. For example you may have noticed that Nescafe, offer its coffee powder in innovative containers which later they can be used as show piece and can be proudly be show cased in the drawing room. The power of good packaging in prompting on the spot purchases cannot be over estimated. Packaging has, thus, emerged as an immensely powerful tool in the consumer goods industry.

The Label

The label is part and parcel of the products package. A label provides written information about the product. Labelling helps the buyer to comprehend the nature of the product, its distinctive features, its composition, its performance and so on. There are grade labels and descriptive labels. In grade labelling, product classifications are based on standards of quality. Products are classified in A, B, C or 1, 2, 3 categories based on quality. But for branded products, mostly descriptive labelling is used, furnishing detailed information about product attributes and quality. For several products, statutory labelling requirements are laid down and marketers are required to follow the mandates laid down as in case of pharmaceutical industry and health and well products industries as well.

1.6 PRODUCT LEVELS

In planning the product offering, the marketer needs to address the five product levels in the figure below. Each level adds more customer value, and the five constitute the customer value hierarchy.

The Customer Value Hierarchy involves the following five product levels which are explained below:

1. *Core Benefit*

The fundamental level in a product is the **core benefit**: the service or benefit the consumer is really buying. A hotel guest is buying rest and sleep. The purchaser of a drill is buying holes. Marketers must see themselves as benefit providers. While customers view themselves as benefit seekers.

2. *Basic Product*

At the second level, the marketer must turn the core benefit into a **basic product**- a version of the product containing only those attributes or characteristics absolutely necessary for it to function. Thus the basic product for a hotel would include “a bed, bathroom, towels, desk, cupboard and closet.” While a basic product for personal hygiene is a bathing soap irrespective whether branded or unbranded.

3. *Expected Product*

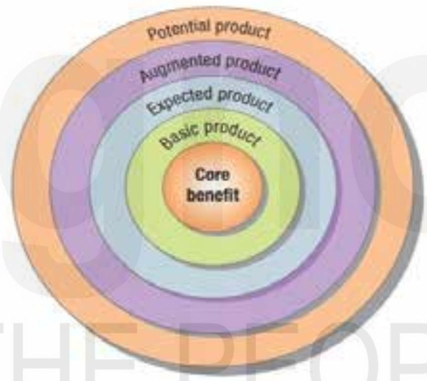
At the third level, the task of every marketer is to prepare an **expected product** - a set of attributes or characteristics that buyers normally expect and agree to when they purchase this product i.e. a clean bed, fresh towels, working lamps and a relative degree of quiet. In developing countries such as India and Brazil, competition takes place mostly at the expected product level. In order to curb competition marketer should nurture and focus at this juncture/stage the customer –value hierarchy accordingly.

4. *Augmented Product*

Constitute the fourth level; the marketer prepares the product through Inclusion of additional features, benefits, attributes or related services that serve to differentiate the product from its competitors. (Brand positioning and competition come in at this level in developed countries)

5. *Potential Product*

At the fifth level stands the potential product.ie all the augmentations and transformations a product might undergo in the future. Here is where companies search for newer ways to satisfy the customer and to differentiate their offerings from the competition. Product innovation comes in at this level for a marketer.



Kotler's 5 Product Levels

Differentiation arises, and competition increasingly occurs, on the basis of product augmentation. Each augmentation adds cost and augmented benefits soon become expected benefits and points-of -parity in the category. If today's hotel guests expect large screen HD TVs, wireless internet access, and a fully equipped fitness centre, the competition must provide even more features to differentiate themselves.

As some companies raise the price of their augmented products, others offer a simplified product for less. Thus you have the Taj Mahal hotel at one end of the spectrum and Ginger hotels at the other.

Activity 1

Imagine you are the Product Manager for a company specializing in air conditioners. Go through all levels of the Customer- Value hierarchy in building up the Potential product for your company. With the space earmarked below show diagrammatically all 5 levels leading up to your potential product.

Space for Diagram for Activity 1

1.5 PRODUCT CLASSIFICATION AND DIFFERENTIATION

Marketers classify products on the basis of durability, tangibility and use (consumer or industrial). Each has an appropriate marketing mix strategy.

1. **Durability and Tangibility: Products fall into three groups according to durability and tangibility:**

   i. Non durable goods are tangible goods normally consumed in one or a few uses, such as soaps and shampoo. Because these goods are purchased frequently, the appropriate strategy is to make them available in many locations, charge only a small markup, and advertise heavily to induce trial and build preference.

   ii. Durable goods are tangible goods, such as refrigerators, machine tools, and clothing, that normally survive many uses. They normally require more personal selling and service, command a higher margin, and require more seller guarantees.

   iii. Services are intangible, inseparable, variable, and perishable products that normally require more quality control, supplier credibility, and adaptability. Examples include haircuts, legal advice and appliance repairs.

2. **Classification of Consumer Goods:**

   i. *Convenience* goods are purchased frequently, immediately, and with minimal effort. Examples include soft drinks,soaps and newspapers.

   Staples are convenience goods consumers purchase on a regular basis.Thus,a consumer might regualarly buy milk, bread, atta etc.

   Impulse goods are goods purchased without any planning or search effort such as Cadbury's chocolate or a magazine at an airport.

   *Emergency good*s are goods purchased when the need is urgent, as an umbrella in the case of unexpected rain. Manufacturers of such goods place them where consumers are likely to experience an urge or a compelling need to purchase.

   ii. *Shopping* goods are those that the consumer characteristically compares on such bases as suitability, quality, price, and style. Examples include furniture, clothing and major appliances.

   iii. *Homogeneous* shopping goods are similar in quality but different enough in price to justify shopping comparisions.Examples would include automobile tyres or a television system.

   iv. *Heterogeneous* shopping goods differ in product features and services that may be more important than price. Sellers of such goods carry a wide assortment to appeal to different tastes and requirements and train their sales people to inform and advise customers appropriately. Examples would be major appliances, clothing, furniture, and high-tech equipment.

   v. *Specialty* goods have unique characteristics or brand identification for which enough buyers are willing to make a special purchasing effort. Example would include cars, men's suits, and perfumes. Specialty goods don't require comparisons.

Buyers only go to dealers with the desired products. The seller only needs to let the prospective customers know where to find them.

vi. *Unsough*t goods are those the consumer does not know about or normally think of buying. Unsought goods require advertising and personal-selling support such as life insurance, smoke detectors, encyclopaedias.

3. Industrial-Goods Classification:

vii. Materials and parts are goods that enter the manufacturer's product completely.

1. Raw materials, in turn, fall into two major groups: farm products and natural products.
2. Manufactured materials and parts fall into two categories: component materials and component parts.

viii. Capital items are long-lasting goods that facilitate developing or managing the finished product.

1. They fall into two groups: installations and equipment.
2. Installations consist of buildings and heavy equipment.
3. Equipment includes portable factory equipment and tools and office equipment.

ix. Supplies and business services are short-term goods and services that facilitate developing or managing the finished product.

1. Supplies are of two kinds: maintenance and repair items and operating supplies.

Business services include maintenance and repair services

Activity 2

Name two convenience goods, two shopping goods that you have used in the recent past. Write below the purpose of your using them and the frequency of your usage of each of these products.

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Why Differentiation?

To be branded, products must be differentiated.

✓ Differentiated products can create significant competitive advantage.

✓ Means for differentiation include form, features, performance quality, conformance quality, durability, reliability, repairability and style.

✓ Design has become an increasingly important differentiator.

Product Differentiation: Need and Scope

Product differentiation can be done through several means:

- Form: the size, shape, or physical structure of a product.

- Features: a company can identify and select appropriate new features by surveying recent buyers and then calculating customer value versus company cost for each potential feature and should avoid feature fatigue.
- Performance Quality: Performance level is the level at which the product's primary characteristics operate: low, average, high, or superior.
- Conformance Quality: the degree to which all produced units are identical and meet promised specifications.
- Durability: a measure of the product's expected operating life under natural or stressful conditions is a valued attribute for vehicles, kitchen appliances, and other durable goods.
- Reliability: a measure of the probability that a product will not malfunction or fail within a specified time period, and high reliability can garner a price premium.
- Repairability: ease of fixing a product when it malfunctions or fails.
- Style: the product's look and feel to the buyer, which creates distinctiveness that, is hard to copy.
- Customization: customized products and marketing allow firms to be highly relevant and differentiated by finding out exactly what a person wants—and doesn't want—and delivering on that.

Services Differentiation:

When the physical product cannot be easily differentiated, the key to competitive success may lie in providing value added services and improving their quality. The main service differentiators are ordering ease, delivery, installation, customer training, customer consulting, maintenance and repair, and ease of returns.

a. *Ordering Ease:* describes how easy it is for the customer to place an order with the company.

b. *Delivery:* refers to how well the product or service is brought to the customer, including speed, accuracy, and care throughout the process.

c. *Installation*: refers to the work done to make a product operational in its planned location.

d. *Customer training:* helps the customer's employees use the vendor's equipment properly and efficiently.

e. *Customer consulting:* includes data, information systems, and advice services the seller offers to buyers.

f. *Maintenance and Repair:* critical in business-to-business settings

g. *Returns:* Avoiding inconvenience, embarrassment , or difficulty in using or handling ; bad for providers when returned merchandise is not in re-sellable condition, or lacks proper proof of purchase, or is returned to the wrong store

a. *Controllable* returns result from problems or errors made by the seller or customer and can mostly be eliminated with improved handling or storage, better packaging, and improved transportation and forward logistics by the seller or its supply chain partners.

b. *Uncontrollable* returns result from the need for customers to actually see, try, or experience products in person to determine suitability and can't be eliminated by the company in the short run.

c. *Basic strategy* is to eliminate the root causes of controllable returns, while developing processes for handling uncontrollable returns.

Product Design: Importance and issues

As competition intensifies design, the totality of features that affect the way the product looks, feels, and functions, offers a potent way to differentiate and position a company's products and services.

- Design Leaders: Use the emotional power of design and the importance to consumers of look and feel as well as function. Hence, design is exerting a stronger influence in categories where it once played a small role. E.g. Jaguar bathroom fittings.
- Power of Design: In a visually oriented culture, transmitting brand meaning and positioning through design is critical.
- Approaches to Design: a well-designed product is easy to manufacture and distribute. To the customer, it is pleasant to look at and easy to open, install, use, repair, and dispose of. The designer must take all these goals into account.

I. Luxury Products face some unique issues

A. Distinguishing Luxury Brands

i. High priced are often about social status and who a customer was—or perhaps aspired to be.

ii. Luxury for many has become more about style and substance, combining personal pleasure and self-expression

iii. Common denominators of luxury brands are quality and uniqueness

iv. Winning formula for many is craftsmanship, heritage, authenticity, and history, and is often critical to justifying a sometimes extravagant price.

B. Growing Luxury Brands: luxury brands that successfully extended their brands vertically across a range of price points are usually the most immune to economic downturns

i. Clear differentiation must exist between these brands, minimizing the potential for consumer confusion and brand cannibalization.

ii. Each also must live up to the core promise of the parent brand, reducing chances of hurting the parent's image.

iii. Horizontal extensions into new categories can also be tricky for luxury brands.

iv. Much of the growth in luxury brands in recent years has been geographical.

C. Marketing Luxury Brands: Do's and Don'ts

a. Globalization, new technologies, financial crises, shifting consumer cultures, and other forces require luxury brand marketers to be skillful and adept at their brand stewardship to succeed.

b. Luxury brand marketers have been issued the following guidelines:

i. Maintain a premium image for luxury brands; control the image.

ii. Create many intangible brand associations and an aspirational image.

iii. Align all aspects of the marketing program for luxury brands to ensure high-quality products and services and pleasurable purchase and consumption experiences.

iv. Besides brand names, other brand elements—logos, symbols, packaging, signage—can be important drivers of brand equity for luxury products.

v. Secondary associations from linked personalities, events, countries, and other entities can boost luxury-brand equity as well.

vi. Luxury brands must carefully control distribution via a selective channel strategy.

vii. Luxury brands must employ a premium pricing strategy, with strong quality cues and few discounts and markdowns.

viii. Brand architecture for luxury brands must be managed carefully.

ix. Competition for luxury brands must be defined broadly because it often comes from other categories.

x. Luxury brands must legally protect all trademarks and aggressively combat counterfeits.

c. Trend for luxury brands is to wrap personal experiences around the products.

d. Some luxury marketers have struggled to find the appropriate online selling and communication strategies for their brand

e. Success depends on getting the right balance of classic and contemporary imagery and continuity and change in marketing programs and activities

Examples of highly successful brands are Ralph Lauren, Calvin Klein, Giorgio Armani and Gucci.

II. Environmental Issues: many firms are considering ways to reduce the negative environmental consequences of conducting business, and some are changing the manufacture of their products or the ingredients that go into them.

III. Product and Brand Relationships: each product can be related to other products to ensure that a firm is offering and marketing the optimal set of products.

1.6 PRODUCT AND BRAND RELATIONSHIPS

Each product can be related to other products to ensure that the firm is offering and marketing the optimal set of products. Often synergies among them can be found and these can lead to substantial savings in terms of costs and effort for the firm.

The Product Hierarchy

A. The Product Hierarchy: Includes six Levels namely

a. Need family: The core need that underlies the existence of a product family. e.g. Security

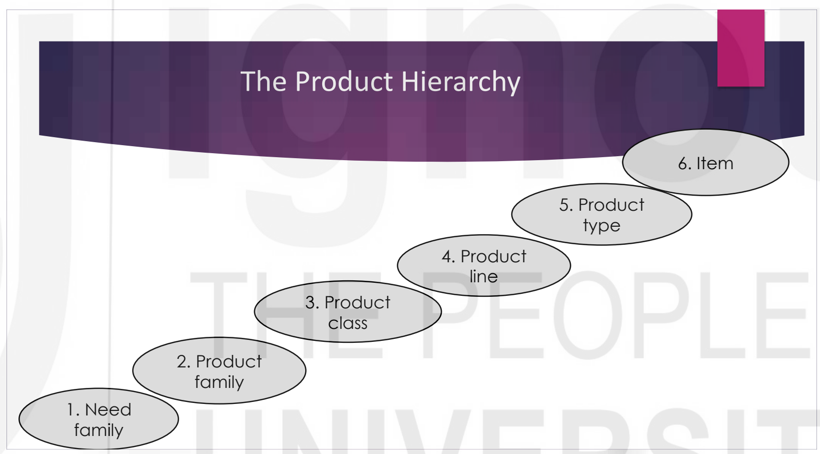
b. Product family: All the product classes that can satisfy a core need with reasonable effectiveness. E.g. savings and income

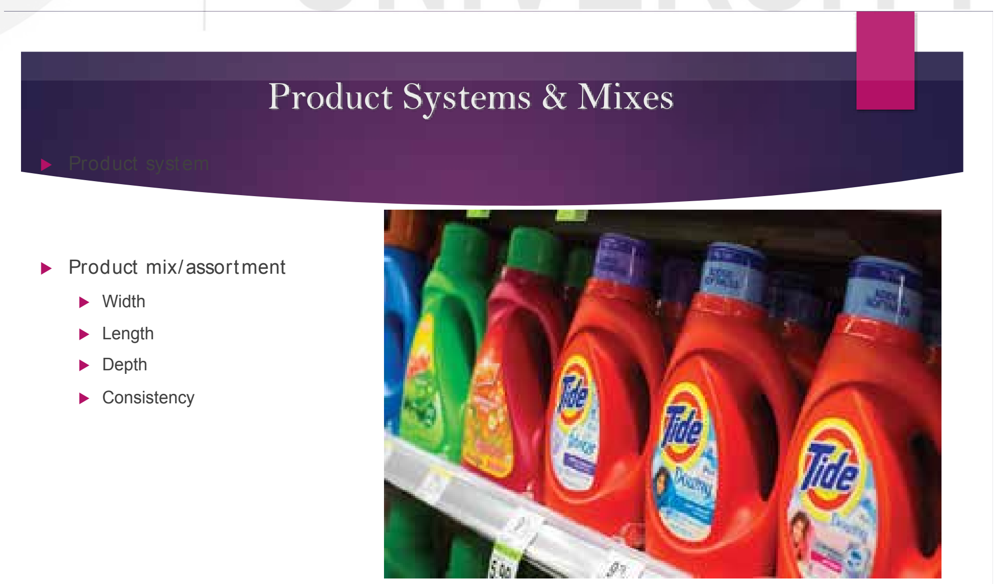
c. Product class: Group of products within the product family recognized as having a certain functional coherence, also known as a product category. e.g. Financial instruments.

d. Product line: group of products within a product class that are closely related because they perform a similar function, are sold to the same customer groups, are marketed through the same outlets or channels, or fall within given price ranges. A product line may consist of different brands, a single family brand, or an individual brand that has been line extended for E.g. Life Insurance.

e. Product type: A group of items within a product line that share one of several possible forms of the product for E.g. Term Life Insurance

f. Item (also called stock-keeping unit or product variant): A distinct unit within a brand or product line distinguishable by size, price, appearance, or some other attribute E.g. Tata AIA Term plan with Return of Premium.

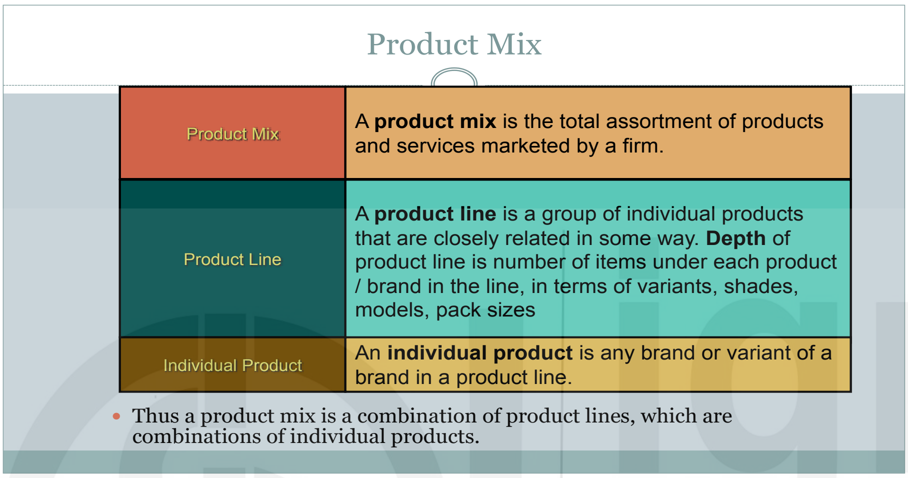


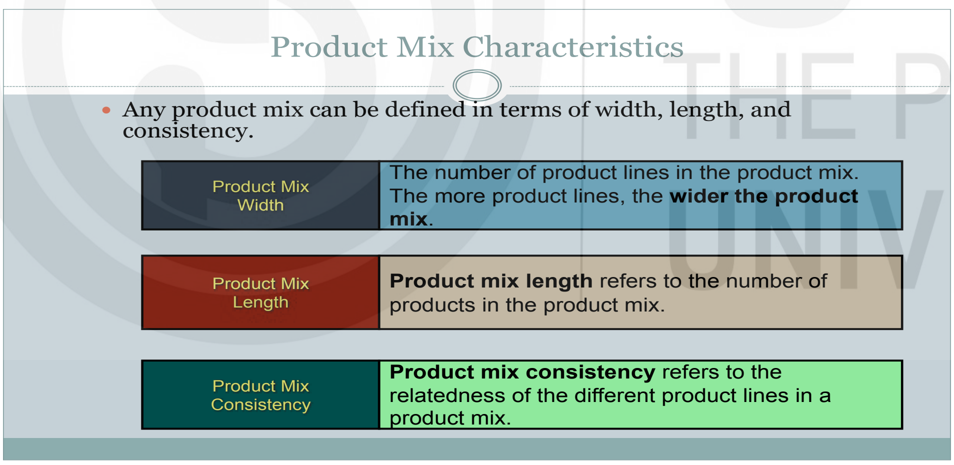


Product Systems and Mixes

A company's product mix has a certain width, length, depth, and consistency.

- The **width** of a product mix refers to how many different product lines the company carries.
- The **length** of a product mix refers to the total number of items in the mix.
- The **depth** of a product mix refers to how many variants are offered of each product in the line.
- The **consistency** of the product mix describes how closely related the various product lines are in end use, production requirements, distribution channels, or some other way





Examples:

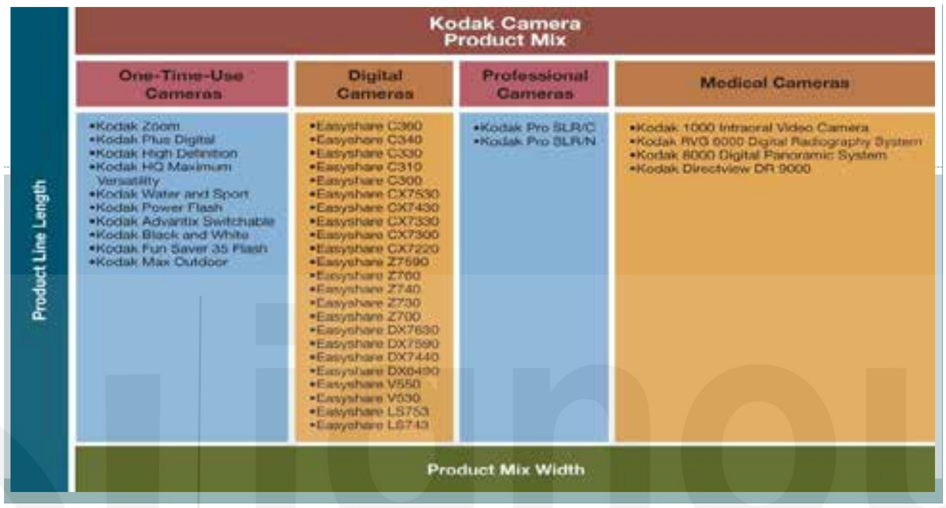
Product mix: - Philips India's Product mix consists of the following product lines, namely music systems and video systems, television, lighting systems and medical electronics catering to different markets across various segments.

Depth of product mix:-Halo shampoo from Colgate-Palmolive comes in three formulations and three sizes and hence has a product mix depth of nine. Similarly Rasna soft drink concentrate from Pioma Industries has seven flavours which form the depth of the product mix.

Consistency of a product:-Philips's product lines are consistent in the sense that they are all electronic products.

These product mix dimensions provide the handles for defining the company's product strategy. The company can increase its business in four ways:

- It can add new product lines, thus widening its product mix. In this way, its new lines build on the company's reputation in its other lines.
- The company can lengthen its existing product lines to become a more full-line company.
- Or it can add more product variants to each product and thus deepen its product mix.
- Finally, the company can pursue more product line consistency—or less—depending on whether it wants to have a strong reputation in a single field or in several fields



Kodak's Product Mix

Line Stretching

As the name suggests, line stretching decisions pertain to adding or dropping items from the line, as and when it is profitable to do so. Line stretching could be upwards, downwards or both ways.

Downward stretch is apt when the company finds its offerings are at a high price end of the market and then stretch their line downwards. For example, Daewoo Motor's, small car "Matiz" of the 90's was originally introduced as a premium high end car with one single variant catering to the top end or the elite segment. Subsequently, the feedback from market and research findings made them offer variants to tap the low end market segment as well. This decision made the company turnaround through a quantum increase in sales and an expansion in customer base.

On the other hand, upward stretch is possible when the company enters the upper end of the market through a line extension. The reason could desire a higher growth rate, increased market share, tapping new market segments, better margins etc. A good example would be that of Lifebuoy soap, which was initially positioned as a hygienic soap focussing on the semi urban and rural mass markets. Currently, Lifebuoy has positioned itself as a premium liquid hand wash, catering to the higher strata of the society.

Reasons for Line Filling: When a firm adds more items within the existing product range results in lengthening the line. Following are the reasons for line filling.

- aiming for incremental profit
- optimal utilisation of excess and under - utilised capacities

- an attempt to offer a full line of the product
- In response to dealer complaints about lost sales because of missing items in the line.

The addition of double toned milk is an example of line filling. Mother Dairy has added this item to its existing range of full cream, and single toned milk, thereby catering to health conscious milk consumers. All the categories of milk mentioned above are easily identifiable by their packaging color.

Activity 3

Select any two examples of line filling in the consumer durables industry. Were the products successful? Give reasons for their success or failure.

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1.7 PRODUCT PLANNING

1. Development of a strategic product planning system is one of the most critical elements of a company's product management function. In designing such plans, the management requires adequate information on the current and anticipated performance of its existing products. This information can again be broadly classified into two dimensions:

   a. the perceptual dimension consisting of the consumers perception about the product per se, as well as in relation to the products of the competitors and

   b. the 'objective' dimension consisting of actual raw information about actual and anticipated performance on relevant criteria such as sales, profits and market share**.**

The information on both the dimensions needs to be seen as a whole to develop a proper product planning system. However, most approaches use the information obtained in isolation, making the picture incomplete. The traditional approaches to Product Planning left much to be desired.

While there are many approaches that companies follow, an integrated approach to product planning has been suggested by Yoram Wind and Henry J. Clayclamp in a paper presented in the Journal of Marketing in 1976. Which is of practical relevance to the marketer are the possibilities they outline at the end of their paper.

- Do not change the product or its marketing strategy;
- Do not change the product but do change its marketing strategy. This may involve a change in the type and level of advertising, distribution and pricing strategies associated with a given positioning and given product attributes.
- Change the product. This may involve product modifications either within the parameters of the product's current market positioning or

within a new positioning. In either case, a change in the associated marketing strategy is required.

- Discontinue the product or the product line. This strategy may involve an interim product or product line "run out" strategy, gradual chopping of the product line, or the immediate phasing out of the product or the complete line.
- Introduce new products into the line or add new product lines.

Which among these to follow would be the outcome of a detailed exercise in Product Planning?

1.8 THE ORIGIN OF THE CONCEPT AND ROLE AND RESPONSIBILITIES OF A PRODUCT MANAGER

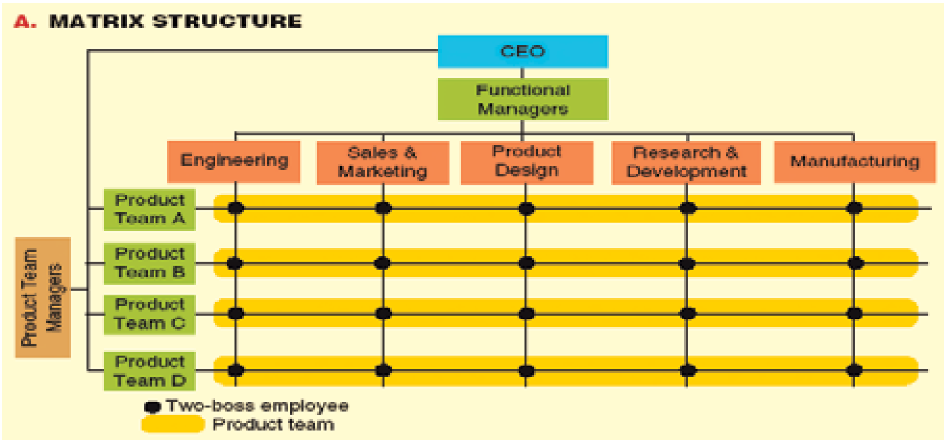
Product Management is the holistic job of Product Managers including planning, forecasting and marketing of products and services. The Product Management concept is widely acknowledged to have been commenced in the year 1931 in Proctor & Gamble. Back then, a manager at P&G named Neil McElroy wrote what is now referred to as the "McElroy Memo". P&G was famous for their culture of writing memos on important topics. McElroy was the manager responsible for Camay soap a lesser brand to the company's leading Ivory soap brand.Camay was not selling well and he decided that a dedicated "brand man" (and supporting team) was needed to ensure that sales of the brand were being maximized.

The need as expressed in his memo was to:

- Understand the regions and volumes of product being shipped.
- For regions where sales are good or growing, understand why and try to apply those principles to other similar regions
- Where sales are light, investigate the situation to understand the problems. Devise a plan to address the problems and work with internal parties to ensure the plan is successful.
- Take charge for all messaging and advertising copy for the brands
- Oversee advertising and marketing expenditures for the brands
- Try new things, particularly with packaging of the brands
- Work with local sales managers to understand their perspective on what is and isn't working in their region.

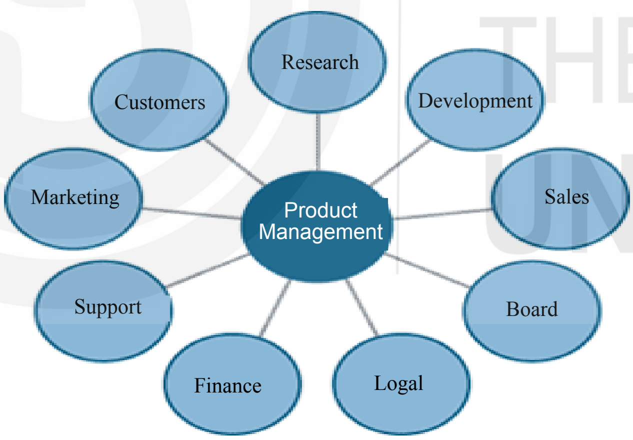
In essence, the "brand man" is responsible for the business success of the brand (product or product family).The role of "brand man" or "brand team" was very successful at P&G and was emulated throughout the consumer packaged goods industry. And, after almost hundred years, Brand Management is well - defined and is a pure marketing and business function within Consumer Packaged Goods (CPG) companies.

Product Management employs a matrix organizational structure in which a product manager is charged with the success of a product or product line, but has no direct authority over the individuals producing and selling the product. Much of the work of a Product Manager is through various departments and cross - functional teams, almost as if the Product Manager were operating a business within a business



Product Manager's reporting in a Typical Matrix Structure

Product Managers, as an organizational form, has moved into a variety of B 2 B firms, as well as service organizations such as financial institutions and hospitals. Most large banks have Product Managers for credit cards, deposit services, trust operations, and commercial cash management services. Product Management is most successful for companies with several products having similar manufacturing but different marketing requirements, particularly when the same product cuts across several divisions or customer groups. Top Management must be committed to the Product Management organization and provide the structure and tools to make it work and the right people must be selected for the job.



Product Manager's Interface

The Product Manager can, thus, be thought of CEO for the Product and responsible for marketing planning, developing product strategy and implementing that strategy through various marketing tools.

1.9 SUMMARY

The Unit takes through and discusses the scope and meaning of Product Management and its ultimate aim of developing, marketing and sustaining new products successfully in the face of competitive rivalry. We have also touched what a product is, its characteristics and classifications, the customer - value hierarchy, means of differentiation, the product hierarchy, products and brand systems, product mix concepts, product and brand relationships, line stretching and filling and the role and responsibilities of a Product Manager as CEO for his brand.

1.10 SELF- ASSESSMENT QUESTIONS

1. Some marketers believe product performance is the main reason for a product to be successful. Other marketers believe look, feel and other design elements are what really make the difference. What do you feel? Take a position on Product performance vs Product aesthetics. (Consider perfumes, paints, Apple I -phones).

2. Imagine you are the Product Manager for a new 5 star hotel. Go through all levels of the Customer- Value hierarchy in building up the potential product for your hotel. What services would you include in the potential product to differentiate yourself from the competition?

3. How important is the packaging for you when you visit the supermarket? What common problems do you face when searching for a product?

   As a Consultant to a luxury soap company what elements / copy would you suggest for them to include in the soap cover?

4. What are the Pros and Cons for Product Management to exist as a separate functional area not falling under the purview of the marketing function?

   Based on your analysis what would be your choice and why?

1.11 REFERENCING/FURTHER READINGS

Ramaswamy & Namakumari, *Marketing Management 6e (2018): Indian Context: Global Perspective,* Sage /Texts.

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Donald R. Lehmann, Russell S. Winer, *Product Management*, McGraw-Hill International Edition.

UNIT 2 PRODUCT LIFE CYCLE (PLC)

Objectives

After reading this unit you should be able to:

- discuss the concept of the product life cycle
- explain the stages in the product life cycle
- operationalise the PLC concept
- understand the product life cycle, its characteristics ,objectives and strategies
- describe the role of PLC in product management decision making
- understand the diffusion of innovation theory and its relevance.

Structure

2.1 INTRODUCTION AND THEORETICAL BACKGROUND

Take a little time away from your busy schedule and look around and you will find for yourself that everything in the market place is presumed to be going through several phases or stages in their life cycle from introduction until its decline. When a product or a brand is launched, it grows, attains maturity, and starts declining to be phased out or may be for a re launch. A Product is basically a need satisfier. In reality there may be several products satisfying the need of a customer. For example, the need for a document and producing multiple documents was existent since human civilization started communication. As civilization matured, the need for documentation has also grown. This change in the level of need is captured by the demand life cycle. For every need cycle there is a sequence of phases starting with emergence, accelerating growth, decelerating growth, maturity, and decline. (The four stages of a product life cycle)

Historically speaking a product is an embodiment of technology, and the technology, in fact, satisfies the need. Let's look at the need for a document was first satisfied by a mud tablet and progressed through palm leaves, copper leaves, paper and has now reached the electronic pages stage. Succeeding technologies normally satisfy the need better than their predecessors. Once you consider the need for multiple copies of the same information, modern technology would suggest use of carbon papers, cyclostyling, photocopying, printing and more convenient and efficient forms of duplication.

The introduction and decline phases of products are common themes in our everyday lives. The cassette tapes have replaced the vinyl records, and currently CDs, digital mini discs, VHS and DVDs tapes have been largely replaced by MP3 digital files. These innovations a result of technology development and technology adoption show that products, markets and competition change over time. Revolutionary products create new product markets. Correspondingly, competitors are developing and copying new ideas and products making existing products out of date more quickly than ever. Think of Nokia and Blackberry and consider how quickly these brands faded in the market. Products, like consumers, go through their respective life cycles.

Roger Everett's Diffusion of innovations theory explains the process of new product adoption over period of time. Both the Product Life Cycle (PLC) and the Diffusion curve are concerned with the changes in the adoption of products over time, the cumulative diffusion curve and PLC curve exhibit considerable similarity. However, there is a difference in terms of the parameters used. PLC depicts absolute sales over time while the diffusion curve relates cumulative percentage of potential consumers over time. Diffusion theory presents a five-stage adoption process, starting from, innovators, early adopters, early majority, late majority, and laggards.

Monopolistic Competition theory explains the variations in the number of firms at different stages of the PLC. Monopolistic competition involves many firms competing against each other, but selling products that are distinctive in some way. Examples include stores that sell different styles of clothing; restaurants or grocery stores that sell different kinds of food; and even products like golf balls or beer that may be at least somewhat similar but differ in public perception because of advertising and brand names.

In the initial stage, the number of firms in the business will be lower, many players would not enter the market due to the risk involved in the new product. Once the product acceptance is realized more firms would enter the market, this leads to the growth in the market. During the maturity stage, the number of firms entering and number of firms leaving the market would be more or less equal. On the decline, only those firms, which choose to specialize, would stay in the market and others would leave the market.

2.2 IMPLICATION OF PRODUCT LIFE CYCLE

A company's positioning and differentiation strategy must change as its product, market and competitors change over the product life cycle. A product life cycle implies the following:

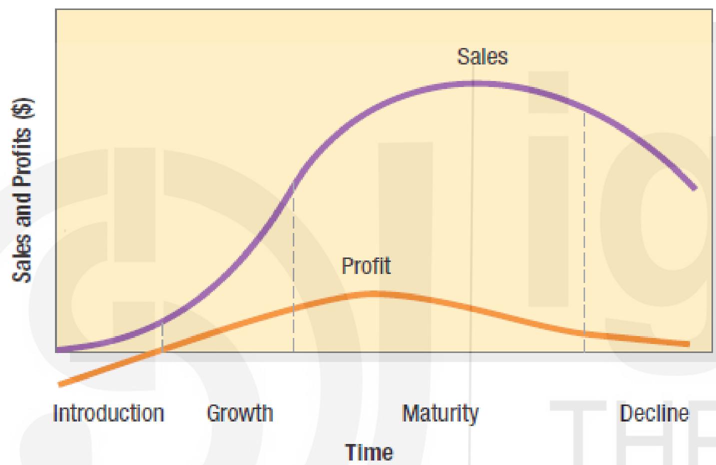
- Products have a limited life
- Product sales pass through distinct stages, each posing different challenges, opportunities, and problems to the seller
- Profits rise and fall at different stages of the product life cycle
- Products require different marketing, financial, manufacturing, purchasing, and human resource strategies in each life-cycle stage<section id="ch11lev2sec7"><title id="ch11lev2sec7.title">

Concept of Product Life Cycle</title>

Most product life cycles are portrayed as bell-shaped curves, typically divided into four stages: introduction, growth, maturity, and decline

- Introduction: A period of slow sales growth as the product is introduced in the market. Profits are non existent because of the heavy expenses of product introduction
- Growth: A period of rapid market acceptance and substantial profit improvement
- Maturity: A slowdown in sales growth because the product has achieved acceptance by most potential buyers. Profits stabilize or decline because of increased competition
- Decline: Sales show a downward drift and profits erode

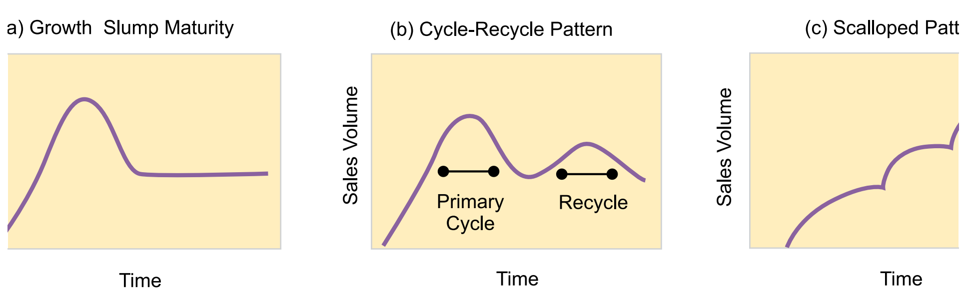
The figure below shows a growth slump maturity pattern, characteristic of small kitchen appliances like bread makers and toaster ovens. Sales grow rapidly when the product is first introduced and then fall to a "petrified" level sustained by late adopters buying the product for the first time and early adopters replacing it.



Source: Philip Kotler Marketing Management 15e

The re- cycle pattern in the figure below describes the usual pattern for new drugs.

The pharmaceutical company aggressively promotes its new drug, producing the first cycle. Later, sales start declining and another promotion push produces a second cycle, usually of smaller magnitude and duration.



Source: *Philip Kotler Marketing Management 15e*

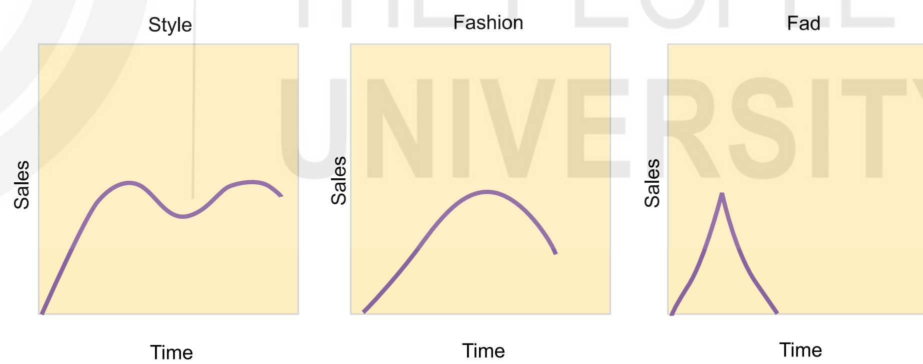
Style, fashion and fad cycles

A style is a basic and distinctive mode of expression appearing in a field of human endeavour and can last for generations and go in and out of vogue.

Thus clothing can be formal, casual or sporty and art realistic, surrealistic or abstract.

- A fashion is a currently accepted or popular style in a given field. Fashions pass through four stages: distinctiveness, emulation, mass fashion, and decline. The length of a fashion cycle is hard to predict. What is currently popular can shift rapidly. Certain colour or style of clothing -baggy jeans, mini -skirts or four inch wide ties may be in fashion on season and outdated the next. Marketing managers who work with fashions have to make really fast fashion changes .For Example, Zara, a fashion retailer headquartered in Spain, takes only about two weeks to go from a new fashion concept to having items on the racks of its stores. One view is that fashions present a compromise and consumers start looking for the missing attributes. For example, if automobiles become smaller and less comfortable, consumers start wanting larger cars. Another view is that if too many consumers adopt the fashion others are discouraged. An alternative view to both the preceding is that the length of a fashion cycle depends on whether the fashion meets a genuine need, is consistent with other trends, satisfies societal norms and values and keeps within technological limits as it develops.
- Fads are fashions that come quickly into public view, are adopted with great zeal, peak early, and decline very fast. Their acceptance cycle is short and they tend to attract only a limited following searching for excitement or wanting to distinguish themselves from others. Examples are fads like the hula hoop, the Rubik's cube, or bell – bottom trousers which came and went.

Fads decline because they don't normally satisfy a strong need. The marketing winners are those recognize fads early a leverage them into products with staying power like Crocs did.



Source: Phillip Kotler Marketing Management 15e

As discussed, a product or brand goes through several phases from birth until its death. Firstly a product is launched; it grows, attains maturity, then starts declining and, finally, is ironed out. Accordingly, the product life cycle is explained by the following stages; (i) Introductory stage; (ii) Growth stage; (iii) Maturity stage and (iv) Decline stage.

Stages in the Product Life Cycle:

1. Introductory Stage

The introductory stage of a product are believed to be relatively slow, even after its technical problems have been ironed out, due to a number of marketing forces and consumer behaviour factors. The major marketing obstacle to rapid introduction of a product is often distribution. Retail outlets are often reluctant to introduce new products, and may prefer to wait until a track record has been established before including them. in their stock.

Consumer acceptance of new products tends to be relatively slow. The newer the product, the greater the marketing effort required to create the demand for it. The length of the introductory period depends on the product's complexity, its degree of newness, its fit into consumer needs, the presence of competitive innovations of one form or another, and the nature, magnitude and effectiveness of the introductory marketing effort.

At this stage it is usually assumed that there are no competitors, the market structure is defined as 'Virtual Monopoly'. But there are very few really radical innovations with no existing substitutes. Most new products and services face considerable competition from existing products, and also experience severe competitive pressure from other new products.

There are many cases where two firms introduce similar products almost at the same time, which is possible if the two companies are working on similar technological developments. On noticing the success of the test market conducted by one company, others may follow suit with similar products. If two or more firms introduce products at about the same time, the result is likely to be a shorter introductory period. The length of the introductory period is a crucial aspect of the PLC. From the managerial point of view, the shorter this stages the better.

The consumers who buy the product in the introductory stage itself are called innovators, and those who buy later are called late adopters or laggards. This may sometimes be misleading, for example in the case of a buyer who hears about a product for the first time two years after its introduction, buying it at once.

2. Growth Stage

The growth stage begins when demand for the new product starts increasing rapidly. If innovators are satisfied with the trial, they move to repeat purchase. They then influence others by word-of-mouth, which is often considered the most effective mode of communication. The product availability and visibility in distribution and in use (e.g., new cars on the road) tend to bring new triers into the market. At this stage, the entry of competitors increases the total demand for the product through their advertising and promotional efforts.

3. Maturity Stage

The maturity or saturation stage occurs when distribution has reached its planned or unplanned peak and the percentage of total population that is ever going to buy the product has been reached. Volume (reflecting the number of customers, quantity purchased, and frequency of purchase) is stable. At this stage it becomes difficult to maintain effective distribution, and price competition is quite common.

4. Decline Stage

Changes in competitive activities, consumer preferences, product technology and other environmental forces tend to lead to the decline of most mature products. If decline is for a product and not brand, producers may withdraw from that product category. The typical reason for a product decline is the entry of new products and decreased consumer interest in the specific product. One of the few options left for keeping a brand alive is price reduction and other drastic means that depress the profit margin and lead to product withdrawal .Product decline occurs when most customers no longer buy the product and only a few loyal customers remain. The latter continue buying the product in spite of no advertising or promotional campaign. The company may decide to follow a `milking or harvesting strategy' i.e., retain the product with meagre marketing support as long as it generates some sales. But this requires maintaining distribution of the product, which becomes less profitable.

Activity 1

1. Trace the PLC of Blackberry cell phones. What were the reasons for their decline?
2. What could they have done to extend the life cycle?

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2.3 PRODUCT LIFE CYLE (PLC) – AN AID TO PRODUCT PLANNING

The Product Life Cycle concept, as a tool of product planning, has had its share of criticism as well as support. Let us look at some empirical evidence in this context.

The S Shaped logistic curve has been widely accepted as a representation of the life cycle of a product, having four visible stages; a long phase (introduction), an exponential phase (growth), a stationary phase (maturity and saturation) and a decline phase.

The empirical support for the S shaped product life cycle is not universal and the evidence regarding its validity is, therefore, not conclusive. This general pattern was found to exist in certain product classes. Puzzle and Cook in their study of 192 consumer products in 1969 found that 52 per cent of the products followed the general pattern of the product life cycle. A number of studies on industrial products also showed that a fairly large percentage of them approximate to the PLC representation.

Sales of `mature' products however, did not necessarily follow the predicted pattern. Three variations in the maturity phase were observed -- the expected stable maturity, a growth maturity due to changes in the market variables, and an innovative maturity due to some innovations introduced in the product. Similarly for the decline stage, it has been shown that the stage is not inevitable for products like milk, bread etc.

It must however, be clearly understood that products do not follow a natural and inevitable cycle of birth, growth and death as organisms do. The product life cycle that they follow is, to a large extent the result, and not the cause, of marketing strategies. The sales and profit graphs do respond to marketing inputs and a sales decline does not necessarily mean that the product has entered an irreversible decline phase. As the life cycle stages do not have predictable duration and inevitable sequence, the product life cycle concept can at best be used as a general guideline for planning future action. It has to be supplemented by a deep and thorough study of the market and competitive conditions characterising a product, in order to serve as a productive or planning tool. The usefulness of the PLC concept varies in different decision situations. As a planning tool, it emphasises the type of main marketing challenges a product is likely to face in the different phases, and suggests major alternative strategies that may be followed at each stage. As a tool for control, the PLC concept enables comparison of product performance against similar products in the past.

2.4 OPERATIONALISING THE PRODUCT LIFE CYCLE

In order to utilize the product life cycle effectively it is necessary to understand how one can unambiguously determine the position of a product in the context of its life cycle. Putting the PLC concept into operation would require the following decisions to be taken:

a) Deciding upon the unit of analysis.

As the actual shape of the product life cycle curve may differ for product form, product class, or an individual product and a brand, it is important to decide which particular unit of analysis is being considered. The PLC analysis can be undertaken for each of these units, both at the firm and the industry level. The importance of explicitly defining a unit of analysis will become apparent to you when you consider an example of whether all TV sets should be considered or the focus should be on either black and white or Colour sets, or portable versus non-portable ones - it is apparent that no set rules can be prescribed at to the correct unit of analysis. Based on their needs and intention of how to use the PLC, Management should define and select the relevant unit of analysis.

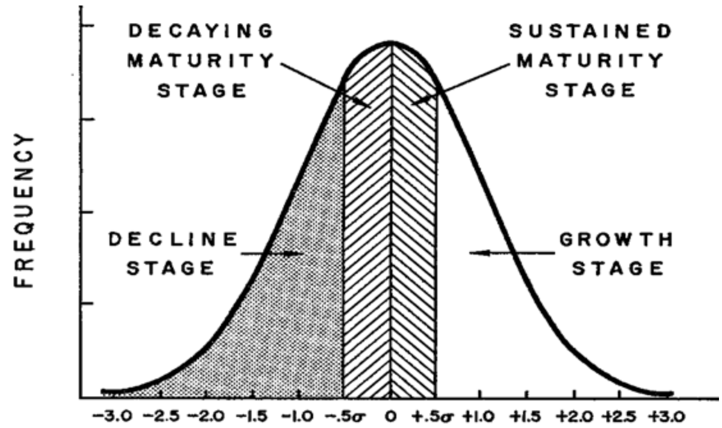
b) Definition of relevant market

Normally, the PLC approach assumes a single homogenous market, which is further segmented by the difference in adoption behaviour of the consumers i.e. the early adopters, late adopters and, finally, the laggards. Generally, PLC studies focus on the product sales at the total market level. However, it is sometimes useful to consider the PLC by type of market (for example international versus domestic market), distribution pattern (direct versus retail distribution) or market segment (organisational buyers versus individual buyers).

c) **Identification of the Product stage in the PLC**

In order to use the PLC concept one would need to answer two related questions (a) how to determine the stage of a product in the life cycle (b) how to determine when the product moves from one stage to another. Since the shape of the PLC curve varies widely for products, as does the duration of the different stages, it is not possible to assess the stage of a product in the life cycle and its transition to another stage, merely by observing the historical sales graph of the product.

Let us consider the operational approach suggested by Polli and Cook as represented in the figure below:



The approach is based on the measurement of the percentage change in real sales from one year to the next. These changes are plotted as a normal distribution. Change in real sales from year to year is then measured. Products having a percentage change less than -.5 are classified as being in the decline stage, those having percentage change greater than .5 were classified as being in the growth stage while those in the range of +.5 were classified as being in the maturity stage, which was further divided into decaying maturity and stable maturity.

As far as the duration of a stage in the PLC of a product and the exact point in time of the transition of a product from one stage to another, no generalised conclusions can be drawn since the duration of each PLC stage depends on a large number of variables like product characteristics, market acceptance and competitive action.

Defining the Unit of Measurement

Though most analyses of PLC are based on actual sales it is important to determine:

1. Whether to use unit sales or rupee sales as unit of measurement?
2. Should actual sales figures or adjusted sales figures be used?
3. Should real or projected prices be used as units of measurement?
4. Should sales be allowed to function as the sole yardstick or should other criteria like profit and market share used to estimate PLC curve?

Determining the time unit

Annual data generally forms the basis of PLC analysis. As the life periods of products are shortening ,and as some classes of products are prone to wide seasonal fluctuations, it may sometimes be considered desirable to use quarterly or monthly data to develop PLC graphs. Depending upon its needs and the product characteristics, management must decide in advance what should be the unit of time chosen.

2.5 PRODUCT LIFE CYCLE -- AS A TOOL TO PLAN MARKET SHARE STRATEGIES

Determining the market share strategy of a product is also one of the key factors in product development. When a product manager is faced with a number of products which are doing well presently he has to come to a decision regarding the long term market share target for each of the products. Given the market share to be obtained, the other elements of the marketing mix can then be decided upon. How do we go about making decisions regarding the market share objectives? What criteria do we use?

A framework to evaluate a product's position vis-a-vis its long term growth prospects has been developed by Bernard Catry and Michel Chevalier. The proposed framework is given below:

Framework Suggested: The first market criterion used is the stage of the product life cycle in which the currently product is. This gives a good indication of the trend in demand as well as competitive patterns. The three phases of a product life cycle used here are introduction, maturity and decline.

The second criterion consists of placing the firm into one of three categories on the basis of its market share. A firm can therefore be placed in (i) small market share (ii) average position (ii) dominant position.

On the basis of these two criteria the firm can choose whether to (i) increase its position (ii) maintain its position at the same level; (iii) reduce its market share.

Table 1 given below indicates the different alternative positions that exist over time in a product group. Each cell corresponds to a different investment or a different expected gain. The term investment in this case means that a firm deciding to increase its position at the introductory stage must make production and marketing investments that will enable it to increase both its brand awareness in its market coverage and its production facilities. On the other hand, a firm which has to maintain its position in a slow growth market may not require as much investment. In the table, the amount of investment necessary is suggested by the number of pluses and minuses following the letter I, while the amount of expected cash return is indicated by pluses and minus following the letter E. An overall value can thus be obtained for each cell by computing the arithmetic sum of cash investments and expected cash returns in the short run. This forms a basis for comparison to arrive at a decision on the appropriate strategy.

Table I : Market share Diagnosis

Product Life Cycle : Market Share Position of the Firm									
Strategic Alternatives	Introduction					Maturity		Decline	
	Small	Average	Dominant	Small	Average	Dominant	Small	Average	Dominant
Increased	I ++	I ++	I +++	I ++	I +++	I ++++	I +	I ++	I ++
Investment	E +++^{1}	E ++$^{(1)}$	E ++$^{(1)}$	E +$^{(1)}$	E +$^{(2)}$	E +$^{(3)}$	E – –$^{(3)}$	E – – –$^{(5)}$	E – – – –$^{(6)}$
Marketing	I	I +	I ++	I	I	I	I +	I	I – –
Position	E^{0}	E + ^{0}	E + + ^{0}	E^{0}	E + + + ^{3}	E + + + + + ^{4}	E +^{2}	E + ^{1}	E + + ^{3}
Disinvestment	I	I–	I–	I–	I– – –	I– – – –	I–	I– – –	I– – –
	E^{0}	E^{1}	E^{0}	E +^{2}	E + + + ^{6}	E + + + + + ^{4}	E^{1}	E + + ^{4}	E + + ^{5}

The action implications arising out of this table can be summarised as under:

1. In the introduction stage of the product life cycle the best short term strategy for a weak brand is to invest in market share. On the other hand, for the dominant firm the best strategy would be to harvest.

2. Firms with a small market position are advised not to invest for more market share at a very late phase of development of the market.

2.6 PRODUCT LIFE CYCLE AS A GUIDELINE FOR MARKETING STRATEGY

In considering the possible relationship between marketing strategy and the product's life cycle, one should see whether companies should adopt their marketing efforts according to the product life cycle or whether the firm's marketing strategy can change the course of the product life cycle. But, unfortunately, most studies of the determinants of product sales are not concerned directly with the product's PLC. Knowledge of the product's stage in its life cycle gives useful but only partial input for the design of the product's marketing strategy.

Recommendations have frequently been made concerning the type and level of advertising, pricing, distribution, and other product/marketing activities required at each of the product life cycle stages. Some of the more common recommendations are discussed below, but should be viewed as hypotheses, and not facts or normative prescriptions.

Advertising:

In the introductory stage, advertising informs customers about the existence, advantages, and uses of new products. During the growth stage -advertising stresses the merits of the products in comparison to competing product. In the maturity phase, advertising attempts to create impressions of product differentiation. It appeals to pride and non-economic utilities. Massive advertising campaigns attempt to attract attention. Finally, in the decline stage, the percentage of sales going into advertising decreases.

Product Changes:

Product changes to be made at each stage of the product life cycle are as follows:

Introduction: new product; Growth: product modification; Maturity: product modification and differentiation; Saturation: product modification, differentiation and diversification; and Decline: product diversification. While at the maturity stage new versions are introduced, at the decline stages, stripped down versions are introduced.

Pricing:

Price is usually believed to be high at the introductory period and to decline with the product life cycle stages, as it becomes an increasingly important competitive weapon, especially at the later stages of growth, maturity and decline. Although price-cutting is quite common in many industries as the product matures (e.g.: Reduction in prices of cell phones, calculators and digital watches over time), many managers prefer to engage in non-price competition. Price cutting in the saturation stage, although common, is far less important than changes in the product, promotion, and distribution policies. In determining the product's price strategy, not only the introductory price should be considered, but also what the next move might be, given alternative competitive actions. A relatively low introductory price should not be ruled out automatically but should be fully examined.

Distribution:

Initial distribution is slow since distribution channels are still being set up to reach its full coverage at the growth stage when retail outlets are seeking the product. At the maturity stage, retail outlets are the first to suffer from changes in consumer purchase patterns; hence manufacturers may start losing outlets. At the same time, efforts are made by manufacturers to establish new methods of distribution e.g. e-commerce and new outlets

The above attempts to prescribe a marketing strategy and guide the allocation of marketing resources over the stages of a product's life cycle assumes that at any one stage of the PLC the firm has only a single reasonable marketing strategy it can flow. This is misleading as it constrains management's creativity in generating new marketing strategies and ignores differences among products, markets and firms. In the real world there is seldom only definitive strategy

Activity 2

Pick up any two products- one each from consumer goods and consumer durables of your choice. Identify the stage of PLC that they are in. As the

Product Manager propose a suitable marketing strategy for products?

Product A...

....

Product B...

....

2.7 PLC: CHARACTERISTICS, OBJECTIVES AND STRATEGIES

TABLE 12.2 Summary of Product Life-Cycle Characteristics, Objectives, and Strategies

	Introduction	Growth	Maturity	Decline
Characteristics				
Sales	Low sales	Rapidly rising sales	Peak sales	Declining sales
Costs	High cost per customer	Average cost per customer	Low cost per customer	Low cost per customer
Profits	Negative	Rising profits	High profits	Declining profits
Customers	Innovators	Early adopters	Middle majority	Laggards
Competitors	Few	Growing number	Stable number beginning to decline	Declining number
Marketing Objectives				
	Create product awareness and trial	Maximize market share	Maximize profit while defending market share	Reduce expenditure and milk the brand
Strategies				
Product	Offer a basic product	Offer product extensions, service, warranty	Diversify brands and items models	Phase out weak products
Price	Charge cost-plus	Price to penetrate market	Price to match or best competitors'	Cut price
Distribution	Build selective distribution	Build intensive distribution	Build more intensive distribution	Go selective: phase out unprofitable outlets
Communications	Build product awareness and trial among early adopters and dealers	Build awareness and interest in the mass market	Stress brand differences and benefits and encourage brand switching	Reduce to minimal level needed to retain hard-core loyals

Source : Phillip Kotler Marketing Management 15e

2.8 EVIDENCE AND CRITIQUE FOR THE LIFE CYCLE CONCEPT

A) Evidence for the Product Life-Cycle Concept

New consumer durables show a distinct take -off, after which sales increase by roughly 45 percent a year, but they also show a distinct slowdown, when sales decline by roughly 15 percent a year. Slowdown occurs at 34 percent penetration on average, well before most households own a new product. The growth stage lasts a little more than eight years and does not seem to shorten over time. Informational cascades exist, meaning people are more likely to adopt over time if others already have, instead of making careful product evaluations. One implication is that product categories with large sales increases at take -off tend to have larger sales declines at slowdown.

B) Critique of the Product Life-Cycle Concept

PLC theory has its share of critics who claim life-cycle patterns are too variable in shape and duration to be generalized and marketers can seldom tell what stage their product is in. As a tool for control, the PLC concept enables comparison of product performance against similar products in the past. It has got a limited utility as a forecasting tool as sales histories of products differ widely and the life cycle stages exhibit varying duration. Further, market evolution takes pace and companies affected by new needs, competitors, technology, channels, and other developments, change product and brand positioning to keep pace. Also like products, markets evolve through four stages: emergence, growth, maturity, and decline, making forecasting uncertain.

2.9 THE DIFFUSION OF INNOVATION THEORY

This is a theory which runs concurrently with the Product life Cycle theory and should be considered complementary in framing of marketing strategy.

Diffusion of Innovation Theory, developed by E.M. Rogers in 1962, is one of the oldest social science theories. It originated in communication to explain how, over time, an idea or product gains momentum and diffuses (or spreads) through a specific population or social system. The end result of this diffusion is that people, as part of a social system, adopt a new idea, behaviour, or product. Adoption means that a person does something differently than what they had done previously (i.e., purchase or use a new product, acquire and perform a new behaviour, etc.). The key to adoption is that the person must perceive the idea, behaviour, or product as new or innovative. It is through this that diffusion is possible.

Researchers have found that people who adopt an innovation early have different characteristics than people who adopt an innovation later. When promoting an innovation to a target population, it is important to understand the characteristics of the target population that will help or hinder adoption of the innovation. There are five established adopter categories, and while the majority of the general population tends to fall in the middle categories, it is still necessary to understand the characteristics of the target population. When promoting an innovation, there are different strategies used to appeal to the different adopter categories.

1. ***Innovators***: These are people who want to be the first to try the innovation. They are venturesome and interested in new ideas. These people are very willing to take risks, and are often the first to develop new ideas. Very little, if anything, needs to be done to appeal to this population.

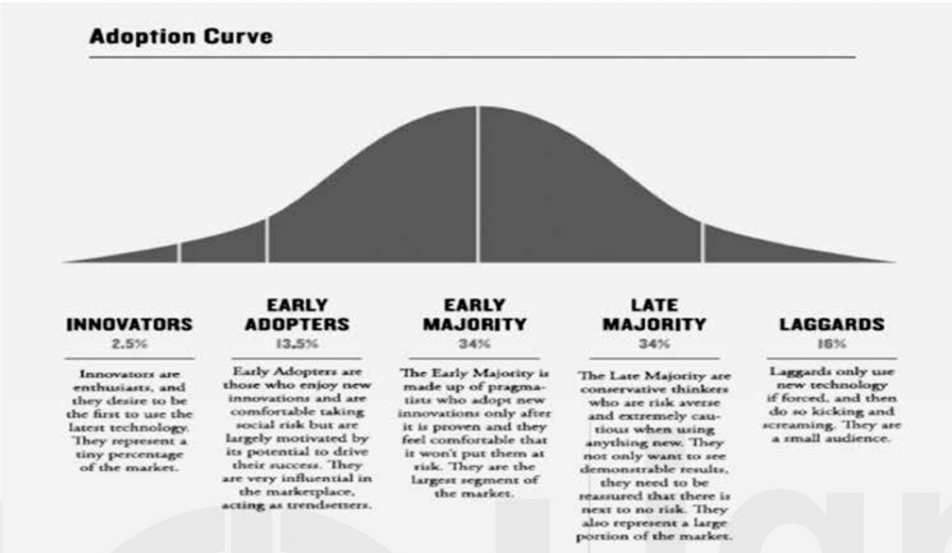
2. ***Early Adopters***: These are people who represent opinion leaders. They enjoy leadership roles, and embrace change opportunities. They are already aware of the need to change and so are very comfortable adopting new ideas. Strategies to appeal to this population include how-to manuals and information sheets on implementation. They do not need information to convince them to change.

3. ***Early Majority***: These people are rarely leaders, but they do adopt new ideas before the average person. That said, they typically need to see evidence that the innovation works before they are willing to adopt it. Strategies to appeal to this population include success stories and evidence of the innovation's effectiveness.

4. ***Late Majority***: These people are sceptical of change, and will only adopt an innovation after it has been tried by the majority. Strategies

to appeal to this population include information on how many other people have tried the innovation and have adopted it successfully.

5. ***Laggards**:* These people are bound by tradition and very conservative. They are very sceptical of change and are the hardest group to bring on board. Strategies to appeal to this population include statistics, fear appeals, and pressure from people in the other adopter groups.



Everett Roger's Diffusion of Innovation Curve

The stages, by which a person adopts an innovation, and whereby diffusion is accomplished, include awareness of the need for an innovation, decision to adopt (or reject) the innovation, initial use of the innovation to test it, and continued use of the innovation. There are **five main factors that influence adoption of an innovation**, and each of these factors is at play to a different extent in the five adopter categories.

1. **Relative Advantage**: The degree to which an innovation is seen as better than the idea, program, or product it replaces.
2. **Compatibility**: How consistent the innovation is with the values, experiences, and needs of the potential adopters.
3. **Complexity**: How difficult the innovation is to understand and/or use.
4. **Trialability**: The extent to which the innovation can be tested or experimented with before a commitment to adopt is made.
5. **Observability:** The extent to which the innovation provides tangible results.

The relevance of this theory would lie in achieving a degree of fair accuracy in forecasting sales and setting sales targets at different stages of the PLC

2.10 SUMMARY

This unit describes the Product life cycle which is one of the enduring and widely publicized frameworks in marketing literature. Both, theory of innovations and diffusion, as well as theory of monopolistic competition complement and endorse this framework. In this model it is conceptualized that product goes through four distinct phases, namely introduction, growth, maturity, and decline. There are controversies about the causality of product life cycle; hence, it is important to take adequate precaution while applying the PLC concept for decision making. PLC can be of diverse shapes, such as the classical bell shaped curve, scalloped, style and so on. The PLC can also be viewed from different levels of product, like core product, product category, brand and so on. The shape may change depending on the markets served. In marketing literature several prescriptions have been proposed for using PLC for marketing strategy formulation. However, it is important to consider the decision context, product in question, market conditions and other factors before using PLC for strategy formulation.

2.11 SELF – ASSESSMENT QUESTIONS

1. Cell phones are a part of our everyday lives. At what stage of the PLC do you think they are? What will marketers have to do to sustain them in future? Or will they die out and be replaced by newer technologies? Think futuristically.
2. Coco Cola and Pepsi have been in a mature stage of the Product Life Cycle for many years. What are these soft drink marketers doing to prevent their decline?
3. Can brands live forever if marketed well or must be they have a finite life? Your views? Trace the history of 3 long - lived brands and find out the reasons for their long term survival.

2.12 REFERENCING/FURTHER READINGS

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UNIT 3 PRODUCT LINE DECISIONS

Objectives

After reading this unit you should be able to:

- explain the concept and logic behind product lines
- discuss reasons for the proliferation of product lines
- enumerate the merits and demerits of product line extensions
- describe the main factors influencing the decision process in relation to product lines
- analyse the changes in product lines of different companies.

Structure

3.1 INTRODUCTION

Across the globe, Products have proliferated at an unprecedented rate in every category of consumer goods and services. There have been widespread line extensions in all types of consumer goods and services. This puts the focus of marketing executives on the product line's length and breadth, the consumers it is catering to, and the future direction to pursue.

A **Product line i**s a group of products which are closely related because they satisfy a class of need or are used together, sold to the same consumer group, marketed through the same types of outlets, fall within similar price ranges, and is considered a unit because of marketing, technical or end-use considerations. In other words, a broad group of products meant for essentially similar uses and which possess reasonably similar physical characteristics constitute a product line. Taking HUL as an example, bath soaps constitute one product line containing Dove, Liril, Pears, Rexona, Lifebuoy and Hamam. Fabric wash is another product line containing Surf, Rin, Wheel, Sunlight, Ala, 501. Barbie's product line contains baby clothes, nursery equipment, vaporizers and toiletries.

3.2 PRODUCT LINE ANALYSIS: - AN INTRODUTION

In offering a product line, companies normally develop a basic platform and modules that can be added to meet different customer requirements and lower production costs. Car manufacturers build cars around a basic platform. Home builders show a model home to which buyers can add additional features. Product line managers need to know the sales and profits of each item in their line to determine which items to build, maintain, harvest, or divest, and they need to understand each product line's market profile and image.

Sales and Profits: Every company's product portfolio contains products with different margins. For instance, supermarkets make almost no margin on bread and milk, reasonable margins on canned and frozen foods and better margins on flowers, ethnic food lines and freshly baked goods. Companies should recognize that different items will allow for different margins and respond differently to changes in level of advertising.

Market Profile and Image: The product line manager must review how the line is positioned against competitors' lines.

A product map shows which competitors' items are competing against the company's items. The map also reveals possible locations for new items. Another benefit of product mapping is that it identifies market segments. Product line analysis provides information for two key decision areas: product line length and product mix pricing.

Product Line length

Company objectives influence product length. One objective is to create a product line by up selling. Thus Maruti would like to move customers from Maruti 800 to Zen to Baleno. A different objective is to create a product line that facilitates cross selling. Hewlett Packard sells printers as well as computers. Still another objective is to create a product line that protects against economic ups and downs. Videocon offers white goods such as refrigerators, washing machines, televisions, microwave ovens, and air conditioners under different brand names to cater to the entry level, mid level, and premium segments. This can also help the company when the economy moves up and down. Companies seeking high market shares and growth will generally carry longer product lines. Companies that emphasize high profitability will carry shorter lines consisting of carefully chosen items.

Product lines tend to lengthen over time. Excess manufacturing capacity puts pressure on the product line manager to develop new items. The sales force and distributors also pressure the company to keep a more complete product line to satisfy customers who like a one -stop shop for all their needs. But as item are added costs rise for design and engineering, inventory carrying, manufacturing changeover, order processing transportation, and new item promotions. Eventually, top management may stop development because of insufficient funds or manufacturing capacity. A pattern of product line growth, followed by massive pruning may repeat itself many times.

Line stretching involves the question of whether a particular line should be extended downwards, upward, or both ways. Line filling raises the question of whether additional items should be added within the present range of the line. Line filling should not lead to cannibalization of the existing products. Line modernization raises the question of whether the line needs a new look and whether the new look should be installed piecemeal or all at once. Line featuring raises the query of the items to feature in promoting the line and which item has to be put in the forefront of promotion to attract customers to that particular category. Line pruning raises the question of detecting and removing weaker product items from the line. It is aimed at getting rid of the dead wood and making the product line more efficient. Deepening is done by adding more items under a given product / brand by way of more variants, models, shades and packs sizes. By enhancing the line depth, more choice is made available to consumers.

Line stretching takes place when a company lengthens its product line beyond its current range, whether down-market, up-market, or both ways. The aim is to enter a new price slot and a new market segment, which is not covered by the existing offers in the line. This can be done in two ways -by line stretching and line filling

Down-Market Stretch: Every company's product line covers a certain part of the total possible range.

A company positioned in the middle market may want to introduce a lower-priced line for any of the three reasons:

- The company may notice strong growth opportunities such as mass retailers like Walmart, Big Bazaar to attract a growing number of shoppers who want value -priced goods.
- The company may wish to foil lower-end competitors who might otherwise try to move up-market.
- The company may find the middle market stagnating or declining.

Examples of successful down – market stretch:

- Surf (now Rs 65/KG), was priced @ Rs 40 per kg for a long time. HUL introduced Sunlight @ Rs 26/Kg and Wheel @ Rs 10/kg primarily to combat the threat posed by Nirma.
- Parker Pen company which historically operated in the high price slot of the pen market as a prestige product, offering low price pens for the mass market.

Marketers face a number of naming choices in deciding to move a brand down-market

- Use the parent brand name on all its offerings.
- Introduce lower-priced offerings using a sub-brand name such as Gillette Vector, the lower- priced version of men's shaving razors.
- Introduce the lower-priced offerings under a different name such as Britannia's Tiger biscuits (expensive to implement wich means brand equity will have to be built from scratch, but the equity of the parent brand name is protected)

The risks associated with down-market stretching

- Not pricing low enough
- Cannibalizing sales
- Consumers not understanding how different offerings are compartmentalized or the differences between them.

Up-Market Stretch: Companies may wish to enter the high end of the market to achieve more growth, realize higher margins, or simply position themselves as full-line manufacturers.

- Some companies have created surprising upscale segments such as Starbucks in coffee, Haagen Daaz in ice – cream and Evian in bottled water.
- Some companies have created new names such as Toyota's Lexus, Nisan's Infiniti and Honda's Acura to counter expected customer resistance.

- till other companies have used their own names in moving up market such as General Electric using the GE Profile brand for its large appliances in the upscale market.

Examples of successful up – market stretch:

Earlier, in the stereo music player market, Philips was synonymous with low priced stereo systems in the Rs 1000 – Rs 2000 range. It launched the Powerhouse range: Rs 6000 Rs 9000. (Catering to the middle segment). Subsequently, it has launched the Power play Range with pricing between Rs Rs.15000 – Rs.25000 (Premium segment) contributing to 30 % of the company's audio sales and succeeded in becoming established as a major player in the upper segment by the time new entrants like Sony arrived Some brands have used modifiers to signal a quality improvement such as Ultra dry Pampers and Extra strength Tylenol.

Two-Way Stretch: companies serving the middle market might stretch their line in both

directions.

***Examples of successful Two -Way stretch*:**

- Haier: Entered with a Refrigerator in the middle range (Rs 20000 –25000). Later, it introduced lower end 170 L models starting from Rs.8990 and high end models 688 L @ Rs.97,900 and above.
- LG also opted for two ways stretching after commencing its launches somewhere in the mid range. In the refrigerator line it went to the lower end with models like LG Economy Direct Cool (55 l) at Rs 6000 and also to the top end with models like LG Premium (816 l) at Rs 1, 55,000 respectively.
- Samsung, too, resorted to two way stretching and finally carried a full line, ranging from the low- priced Direct Cool (170 l) at Rs 9000 to Samsung (570 l) at Rs 78.500.

Line Filling: In line filling, the firm introduces more items to the line to strengthen the gaps in its current range of offers. The intention is that it should be seen by consumers as a "full line" company, and consumers do not go to competitors for particular variants /models. If overdone, it can result in cannibalization and confusion.

***Example of successful Line Filling*:**

- Good Knight Mosquito repellents which have been successful .The brand have been offered in different product forms and variants depending upon consumer requirements.
- Daimler India launched a medium duty truck in the sub 9 tonne category. The company had a gap in this segment. With this addition, it became a full line truck company with products ranging from 6 tonnes to 49 tonnes.

Line Modernization Featuring and Pruning: Product lines regularly need to be modernized.

- A piecemeal approach allows the company to see how customers and dealers take to the new style and is less draining on the company's cash flow.
- The downside is it that lets competitors see its changes and they start redesigning their own lines.

- Sometimes a company finds one end of its line selling well and the other end selling poorly.
- Using sales and cost analysis, product line managers must periodically review the line for deadwood that depresses profits.
- Multi-brand companies all over the world try to optimize their brand portfolios.

In rapidly changing markets modernization is continuous. Companies plan improvements to encourage consumers to migrate to higher valued, higher priced products. Microprocessor companies such as Intel and AMD continually introduce more advanced versions of their products.

In the Indian market there are more than five thousand models with leader Titan watches offering as many as 1500 of these. Titan makes 350 - 400 new launches annually. In mobile phones, the top 10 companies launch nearly 400 models in a single year. The same is the practice in products ranging from high priced durables like cars and low – priced daily consumption items like soaps.

Pruning

Managing too long a product line with too many forms can cause problems. For optimising the profits of a product line, it must be manageable in the first place. An analysis of the sales trends of all items in the portfolio needs to be done regularly. The ones that are uneconomic, or cannibalise the others, need to be weeded out.

In 1999, after realizing more than 90 % of its profits came from just 400 brands, Unilever did a massive weeding programme to get the most value from its brand portfolio by eliminating three quarters of its 1600 distinct brands by 2003.

A multi -brand company P& G, in its "back to basics" strategy concentrated on brands over $ 1 billion in revenue such as Tide, Crest, Pampers and Pringles.

While Titan introduces 350 -400 models each year, in different segments ,ranging from kids watches in the Rs 300 range to jewellery watches costing lakhs of rupees, each year it also withdraws an equal number of models, so that the line remains modern, manageable and profitable.

Deepening the Product Lines

Variants are added to the main brand to satisfy the consumer's need for variety. They also help push visibility of the brand at the store level. Colours, fragrances, tastes, flavours, are all used to introduce variants and increase their appeal. Lux bath soap was available initially only in the colour white. Now it is available in different colours /flavours/bases/, and each in different size.

Activity 1

Ascertain the product lines of the top 3 detergent (e.g. HUL, P&G and Nirma) manufacturers in India. Suggest which products they should drop from their product lines and why?

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3.3 BASIS FOR PRODUCT LINE EXTENSION

The following seven important factors make companies pursue line extensions as a significant element of their marketing strategies.

Customer Segmentation

Line extension is perceived by managers as a low-cost, low-risk way to meet the needs of various customer segments, and by using more sophisticated and lower-cost market research and direct-marketing techniques; they can identify and target finer segments more effectively than ever before. In addition, the depth of audience-profile information for television, radio, and print media has improved; managers can now translate complex segmentation schemes into efficient advertising plans.

Consumer Desires

There is widespread volatility in consumer behaviour and brand loyalty has taken more or less a back seat in many different categories of consumer goods and services. More consumers than ever are switching brands and trying products they have never used before. Line extensions try to satisfy the desire for "something different" by providing a wide variety of goods under a single brand umbrella. Such extensions, companies hope, would fulfil customers' desires while keeping them loyal to the brand franchise. Moreover, according to studies conducted by the Point-of-Purchase Advertising Institute, USA, consumers now make around two-thirds of their purchase decisions about grocery and health and beauty products on impulse while they are in the store. Line extensions, if stocked by the retailer, can help a brand increase its share of shelf space, thus attracting consumer attention. When marketers coordinate the packaging and labelling across all items in a brand line, they can achieve an attention-getting billboard effect on the store shelf or display stand, and, thus, leverage the brand's equity.

Pricing Breadth

Managers have found a novel way of increasing profitability through line extension. Managers often tout the superior quality of extensions and set higher prices for these offerings than for core items. In markets subject to slow volume growth, marketers can then increase unit profitability by trading current customers up to these "premium" products. In this way, even cannibalized sales are profitable at least in the short run.

In a similar spirit, some line extensions are priced lower than the lead product. For example, American Express offers the Optima card for a lower annual fee than its standard card, and Marriott introduced the hotel chain Courtyard to provide a lower-priced alternative to its standard hotels. Extensions give marketers the opportunity to offer a broader range of price point's in order to capture a wide audience.

Excess Capacity

Line extension helps in utilizing the excess capacity of the production facilities of the firm. Many manufacturing operations add faster production lines to improve efficiency and quality. The same organizations, however, do not necessarily retire existing production lines. The resulting excess capacity encourages the introduction of line extensions that require only minor adaptations to current products.

Short-Term Gain

Besides sales promotions, line extensions represent the most effective and least imaginative way to increase sales quickly and inexpensively. The development time and costs of line extensions are far more predictable than they are for new brands, and less cross-functional integration is required.

In fact, few brand managers are willing to invest the time or assume the career risk to introduce new brands to market. Line extensions offer quick rewards with minimal risk. Senior executives often set objectives for the percentages of future sales to come from products recently introduced. At the same time, under pressure from stock exchanges for quarterly earnings increases, they do not invest enough in the long- term research and development needed to create genuinely new products. Such actions necessarily encourage line extensions.

Competitive Intensity

Mindful of the link between market share and profitability, managers often see extensions as a short-term competitive device that increases a brand's control over limited retail shelf space and its overall demand for the category. Close-up and Colgate toothpastes brands both are available in more than 15 types and package sizes, have increased their market shares in the last decade at the expense of smaller brands that have not been able to keep pace with their new offerings.

Trade Pressure

The proliferation of different retail channels for consumer products, from club stores to hypermarkets, pressures manufacturers to offer broad and varied product lines. While retailers object to the proliferation of marginally differentiated and "me-too" line extensions, trade accounts themselves contribute to stock-keeping unit (SKU) proliferation by demanding either special package sizes to fit their particular marketing strategies (for example, bulk packages or multipacks for low-price club stores) or customized, derivative models that impede comparison shopping by consumers.

3.4 THE DISADVANTAGES OF LINE EXTENSIONS

Given this backdrop, it's easy to visualise why so many managers have been swept into line-extension mania. But, as more managers are discovering, the problems and risks associated with extension proliferation are formidable.

Weaker Line Logic

Manager often extend a line without removing any existing items. As a result, the line may expand, to the point of over segmentation, and the strategic role of each item becomes muddled. Salespeople should be able to explain the commercial logic for each item. If they cannot, retailers turn to their own data the information collected by checkout scanners to help them decide which items to stock. Invariably, fewer retailers stock an entire line. As a result, manufacturers lose control of the presentation of their lines at the point of sale, and the chance that a consumer's preferred size or flavour will be out of stock, increases.

A disorganized product line can also confuse consumers, motivating those less interested in the category to seek out a simple, all-purpose product, such as All Temperature Cheer in the Laundry detergent category.

Lower Brand Loyalty

Some marketers mistakenly believe that loyalty is an attitude instead of understanding that loyalty is the behaviour of purchasing the same product repeatedly. In the past 50 years, many of the oldest and strongest brands have had two and three generations of customers buying and using products in the same way. When a company extends its line, it risks disrupting the patterns and habits that underlie brand loyalty and re-opening the entire purchase decision.

Although line extensions can help a single brand satisfy a consumer's diverse needs, they can also motivate customers to seek variety and, hence, indirectly encourage brand switching. In the short run, line extensions may increase the market share of the overall brand franchise. But if cannibalization and a shift in marketing support decrease, the share held by the lead product and the long-term health of the brand will be weakened. This is particularly true when line extensions diffuse rather than reinforce a brand's image in the eyes of long-standing consumers, without attracting new customers.

Underexploited Idea

By bringing important new products to market as line extensions, many companies leave money on the table. Some product ideas are big enough to warrant a new brand. The line extensions serve the career goals of a manager of an existing brand better than a new brand does, but long-term profits are often sacrificed in favour of short- term risk management.

Stagnant Category Demand

Line extensions rarely expand total category demand. People do not eat or drink more, wash their hair more, or brush their teeth more frequently, simply because they have more products to choose from. In fact, a review of several product categories shows no positive correlation between category growth and line extensions. If anything, there is an inverse correlation as marketers try in vain to reinvigorate declining categories and protect their shelf space through insignificant line extensions.

Poorer Trade Relations

Retailers cannot provide more shelf space to a category simply because there are more products within it. They have responded to the flood by rationing their shelf space, stocking slow-moving items only when promoted by their manufacturers, and charging manufacturers slotting fees to obtain shelf space for new items and failure fees for items that do not meet target sales within two or three months. As a manufacturer's credibility has declined, retailers have allocated more shelf space to their own private label products. Competition among manufacturers for the limited slots still available escalates overall promotion expenditures and shifts margins to the increasingly powerful retailers.

More Competitor Opportunities

Share gains from line extensions are typically short-lived. New products can be matched quickly by competitors. What's more, line-extension proliferation reduced the retailer's average turnover rate and profit per SKU (Stock Keeping Unit) this can expose market leaders to brands that do not attempt to match all the leaders' line extensions but instead offer product lines concentrated on the most popular line extensions. As a result, on a per - SKU basis, lesser known brands, as compared to market leaders, can deliver a higher direct product profit to the retailer than brands with -larger shares and more SKUs.

Activity 2

Take two leading Washing Machine manufacturers (e.g. Siemens and IFB) in India and analyse the changes in their product line over the last five years and its effect on the total market share of each company.

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3.5 FACTORS INFLUENCING PRODUCT LINE DECISIONS

An in-depth analysis of a product's potential to achieve a desired level of return on the company's investment is an essential component of the marketing planning process. An analysis of this type not only assesses financial opportunities but also provides ideas about how to compete better given structural characteristics of the category.

The various factors influencing the product line decisions of a corporation are:

Category Size

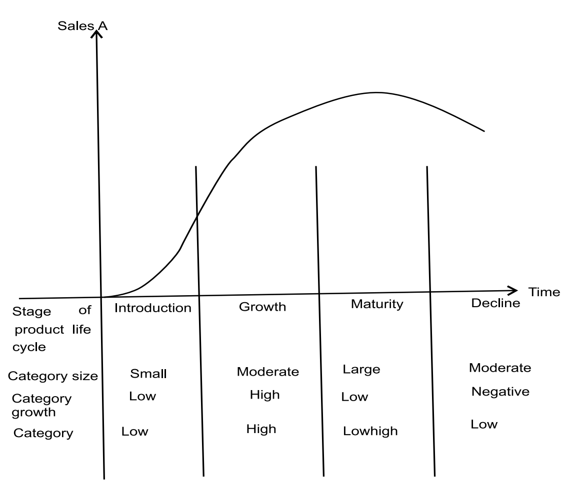
Category size is an important piece of data about any market. It is clearly an important determinant of the likelihood that a product will generate revenues to support a given investment. In general, larger markets are better than smaller ones. Besides having more market potential, large categories usually offer more opportunities for segmentation than small ones. Large markets, however, tend to draw competitors with considerable resources, thus making them unattractive for small firms.

Market Growth

Market growth is a key market factor advocated by various planning models. Not only is current growth important, but growth projections over the horizon of the plan are also critical. Fast-growing categories are almost universally desired due to their abilities to support high margins and sustain profits in future years. However, like large categories, fast-growing ones also attract competitors.

Product Life Cycle

Category size and category growth are often portrayed simultaneously in the form of the product life cycle. Usually presumed to be S shaped, this curve breaks down product sales into four segments: introduction, growth, maturity, and decline. The introduction and growth phase are the early phases of the life cycle when sales are growing rapidly, maturity represents a levelling off in sales, and the decline phase represents the end of the life cycle.



Category Attractiveness over the Product Life Cycle

In the introductory phase, both the growth rate and the size of the market are low, thus making it unattractive for most prospective participants, who would rather wait on the side-lines for a period of time. When market growth and sales start to take off, the market becomes more attractive. In the maturity phase, the assessment is unclear; while the growth rate is low, the market size could be at its peak. Finally, the decline phase usually is so unattractive that most competitors flee the category.

Sales Cyclicity

Many categories experience substantial inter year variation in demand. Highly capital-intensive business such as automobiles, steel and machine tools, are often tied to general business conditions and, therefore, suffer through peaks and valleys of sales.

Seasonality

Seasonality - intra -year cycles in sales is generally not viewed positively. Seasonal business tends to generate price wars because there may be few other opportunities to make substantial sales. However, most products are seasonal to some extent. Some, of course, are exclusively seasonal.

Profits

While profits vary across products or brands in a category, large inter -industry differences also exist.

These differences in profitability across industries are actually based on a variety of underlying factors. Differences can be due to factors of production (e.g., labour versus capital intensity, raw materials), manufacturing technology, and competitive rivalry. Product categories that are chronically low in profitability are less attractive than those that offer higher returns.

A second aspect of profitability is that it varies over time. Variance in profitability is often used as a measure of industry risk. Product managers must make a risk-return trade-off, evaluating the expected returns against the variability in those returns.

Attractiveness of Market Variables

	Attractiveness	
	High	Low
Market size	+	–
Market growth	+	–
Sales cyclicity	–	+
Sales seasonality	–	+
Profit level	+	–
Profit variability	–	+

3.6 CATEGORY FACTORS INFLUENCING PRODUCT LINE DECISIONS

Although the aggregate factors just described are important indicators of the attractiveness of a product category, they do not provide information about underlying structural factors in assessing the structure of industries: such as

- The threat of new entrants.
- The bargaining power of buyers
- The bargaining power of suppliers.
- The amount of intra -category rivalry.
- The threat of substitute products or services.

Threat of New Entrants

If the threat of new entrants into the product category is high, the attractiveness of the category is diminished. Except for the early stages of market development, when new entrants can help a market to expand, new entrants bring additional capacity and resources that usually heighten the competitiveness of the market and diminish profit margins. Even at early stages of market growth, the enthusiasm with which new entrants are greeted is tempered by who the competitors are .The strength of barriers to entry erected by the existing competition impact the likelihood of new competitors entering the market. Some of the potential barriers to entry are -

- Economies of Scale
- Product Differentiation
- Capital Requirements
- Switching Costs
- Distribution

Bargaining Power of Buyers

Buyers are people or institutions that receive finished goods or services from the organizations in the category being analysed. Buyers can be distributor's manufacturers, original equipment manufacturers (OEMs), or end - customers.

High buyer bargaining power is negatively related to industry attractiveness. In such circumstances, buyers can force down prices and play competitors off against one another for benefits such as service. Some conditions that occur when buyer bargaining power is high include the following:

- When the product bought is a large percentage of the buyer's costs.
- When the product bought is undifferentiated.
- When the buyers earn low profits.
- When the buyer threatens to backward integrate.
- When the buyer has full information.
- When substitutes exist for the seller's product or service.

Again, the product manager's concern is to decrease the buying power. This can be accomplished, by increasing product differentiation (e.g., making services such as technical assistance or manufacturing-related consulting built into switching costs).

Bargaining Power of Suppliers

Suppliers are any institutions that supply the category of concern with factors of production such as labour, capital raw materials, and machinery.

High supplier's power is clearly not an attractive situation because it allows suppliers to dictate price and other terms, such as delivery dates, to the buying category. Some conditions that prevail when supplier bargaining power is high are:

- When suppliers are highly concentrated, that is, dominated by a few firms.
- When there is no substitute for the product supplied.
- When the supplier has differentiated its product or built in switching costs.
- When supply is limited.

Current Category Rivalry

Product categories characterized by intense competition among the major participants are not as attractive as those in which the rivalry is more sedate. A high degree of rivalry can result in escalated marketing expenditures, price wars, employee raids, and related activities. Such actions can exceed what is considered `normal' market competition and can result in decreased welfare for both consumers and competition. Some of the major characteristics of categories exhibiting intensive rivalries are:

- Many competitors
- Slow growth
- High fixed costs
- Lack of product differentiation
- Personal rivalries
- Pressure from substitutes

Categories making products or delivering services for which there are a large number of substitutes are less attractive than those that deliver a relatively proprietary product, one that uniquely fills a customer need or solves a problem. Since almost all categories suffer from the availability of substitutes, this may not, be a determinant of an unattractive product category. However, some of the highest rates of return are earned in categories in which the range of substitutes is small.

Category Capacity

Constant overcapacity is not a positive sign for long-term profitability. When a category is operating at capacity, its costs stay low and its bargaining power with buyers is normally high. Thus, a key indicator of the health of a category is whether there is a consistent tendency toward operating at or under capacity.

Impact of Category Factors on Attractiveness

	High	Low
Threat of new entrants	–	+
Power of buyers	–	+
Power of suppliers	–	+
Rivalry	–	+
Pressure from substitutes	–	+
Unused capacity	–	+

Activity 3

If you are a European fashion clothes manufacturer aiming at the Indian market, what are the conditions that will influence your product line decisions? Outline all the factors and decide upon the length of your product line.

....
....
....
....
....
....

3.7 SUMMARY

As discussed there are multiple external and internal factors affecting the product line decisions to be made by the Product Manager? Product Line decisions are ultimately a function of the company's short term and long term objectives and the external and internal factors prevailing in the environment and within the company. This is the only basis for reaching at a product line decision aimed at improving the overall profitability of the firm.

3.8 SELF ASSESSMENT QUESTIONS

1. Undertake a study to find out the changes in LG's different product lines (ACs and Refrigerators) in the last decade i.e. 2010 and study the impact on their market shares. Were their strategies successful? Why or why not?
2. HUL has a large number of product lines. Is it right to have so many or would it be too much of a strain on its resources in future to maintain them?

3.9 REFERENCES/ FURTHER READINGS

Ramaswamy & Namakumari, *Marketing Management 6e (2018) : Indian Context: Global Perspective,* Sage /Texts.

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UNIT 4 PRODUCT PORTFOLIO

Objectives

After reading this unit you should be able to:

- explain the concept of product portfolio analysis
- describe the various types of product portfolio models
- discuss the usage of portfolio models for decision making.

Structure

4.1 INTRODUCTION

Product portfolio analysis is an increasingly popular framework for strategic planning within multi-product corporations**.** Unless the entire product portfolio is considered explicitly for decision making, sub optimal decisions are quite likely. In product portfolio analysis, all the products of the company are evaluated on important dimensions like, profitability, growth potential and associated risk involved in investment etc. Such an evaluation facilitates and provides valuable inputs for deciding on investment on product market strategies like, addition of products, modification of existing products, and deletion of low performing products.

4.2 PORTFOLIO CONCEPT

The basis of the portfolio concept lies in management of financial resources wherein the optimal portfolio of stocks are decided based on the trade off between the expected returns and associated risk of the stock.

Peter Drucker has suggested a six-fold product classification-for effective resource allocation. His classification tends to have two product groups with positive contributions one with a marginal contribution and three with negative contributions. A description of these groups follows:

1. Tomorrow's breadwinners
2. Today's breadwinners

3. Products with potential to contribute if attended to yesterday's breadwinners
4. Yesterday's breadwinners
5. "Also ran"
6. Failures

In this typology, resource allocation is based on the ranking. Tomorrow's breadwinners and today's breadwinners are provided with necessary resources. While yesterday's bread winners "Also ran" and Failures are deprived of resources and sometimes allowed to die. The products falling at rank three are provided with resources depending on their expected potential.

In this procedure, product contribution margin is used as the sole decision making indicator. Subsequently, more sophisticated portfolio models have been developed incorporating several criteria like, market share, market growth, profitability, expected return, risk etc.

The advantage of product portfolio analysis is that it offers a structured set of dimensions to evaluate the current products comprehensively. The dimensions may vary among different portfolio models proposed in the marketing literature. There are two kinds of portfolio models the first suggests normative dimensions as in the case of the share/growth matrix proposed by the Boston Consulting Group, popularly known as the BCG matrix comprising relative market share and industry growth rates as the two dimensions.

The primary tasks involved in portfolio analysis are to classify the products on the dimensions of the matrix and to develop strategies to move the products on the matrix towards a target or ideal matrix. The target matrix may include potential segments, products etc. This extended portfolio would reflect the objectives of the management, and the desired direction of growth. This would help in devising guidelines for resource allocation.

For constructing the product portfolio matrix, three critical decisions have to be considered as mentioned below:

- trend of the business considered for analysis,
- defining the served markets
- time frame for analysis.

Level of Business Unit

Product portfolios could be analysed at different levels starting from Corporate, Strategic Business Units, product lines and so on, so forth.

Served Market

Product portfolios could be constructed for every segment as well as across served market segments. The served market refers to the market segments of the total market within which a firm actively competes. Quite often portfolio analysis is carried out not only with respect to the served market, but for higher level product markets as well. They are, in fact, complementary to each other. When the portfolio is restricted to the served market, the focus is on the consumer point of view, such as positioning and consumer perception about the complete assortment provided to the market. At a higher level portfolio analysis the focus is more on performance of each product in the served market.

Time Frame for Analysis

The first dilemma the decision maker encounters is whether to use historical data or to incorporate the period for which decisions are being taken.

However, mostly historical data is used for the analysis with an assumption that historical trends of growth are going to continue in the decision period, too. In the contexts where this assumption does not hold good, forecasts have to be made for the decision period and incorporated in the portfolio model.

4.3 THE LOGIC UNDERLYING THE PORTFOLIO APPROACH

The portfolio models analysis is based on the following philosophy:

1) The opportunities for the product/market differ.
2) Products inherit different competitive strengths and exploiting opportunities.
3) In resource allocation decision for products, the major considerations are opportunity for product growth and profitability.
4) Corporate objectives would be decided based on the cumulative opportunities for all the products and competitive strength of these products.
5) Based on the corporate and individual product objectives, resources would be allocated. However, it is not a straightforward process; it involves several interactions based on much involved analysis of sources and uses of resource

4.4 TYPES OF DISPLAY MATRICES

The purpose of portfolio analysis is to optimally allocate resources for the best total return, with focus on the corporate strategies. Many different approaches involving different matrices have evolved over the years; with the common objective of successful diversification. We discuss below these matrices:

1. Boston Consulting Group's Growth-Share Matrix

This is a two-by-two product portfolio analysis using Market Growth Rate and Relative Market Share. It identifies its segments as Dogs, Cows, Stars and Question Marks (also called Wild Cat or Problem Children). These different businesses are categorized in terms of cash flow. Each segment is then populated by bubbles whose size is proportional to the size of the business activity, expressed in terms of sales, assets or some other measure.

2. McKinsey Matrix

This matrix is generally associated with General Electric and Shell Companies. It is a three-by-three matrix, which divides Industry Attractiveness and Business Strength, into low, medium, and high segments each. The parameters are compound variables of different factors, to be either subjectively judged or objectively computed based on weighted judgements.

3. Strategic Planning Institute's Matrix

Strategic Planning Institute's program on Profit Impact of Market Strategy (PIMS) compares the company's profitability with the average profitability of the associated industry. DIMS matrix is based on business average profitability (PAR ROI), an industry characteristic determined by a cross-sectional, multi- dimensional regression study of the profitability to different businesses. This method avoids the judgmental weights of the previous approach, but some criticize this approach because of its heterogeneous population of dissimilar businesses.

4. Arthur D. Little Company's Matrix

This matrix uses Life Cycle Stages (Embryonic, Growth, Mature and Decline) with Business Strength (Weak, Tenable, Favourable, Strong and Dominant). The grid segments are then classified into Build, Hold, Harvest and Unacceptable ROI

5. Hofer's Product / Market Evolution Matrix

In this matrix, products are plotted in terms of their product /market evolution and their competitive position.

Besides the above, there are other matrices associated with different consultants who have developed them to suit their specific needs for market differentiation. Some of the above mentioned matrices are described in detail below, starting with the pioneering BCG matrix.

4.5 THE BCG GROWTH-SHARE MATRIX

BCG Portfolio Analysis is based on the premise that majority of the companies carry out multiple business / products / activities in a number of different product-market segments. Together these different businesses form the Business Portfolio, which can be characterized by two parameters:

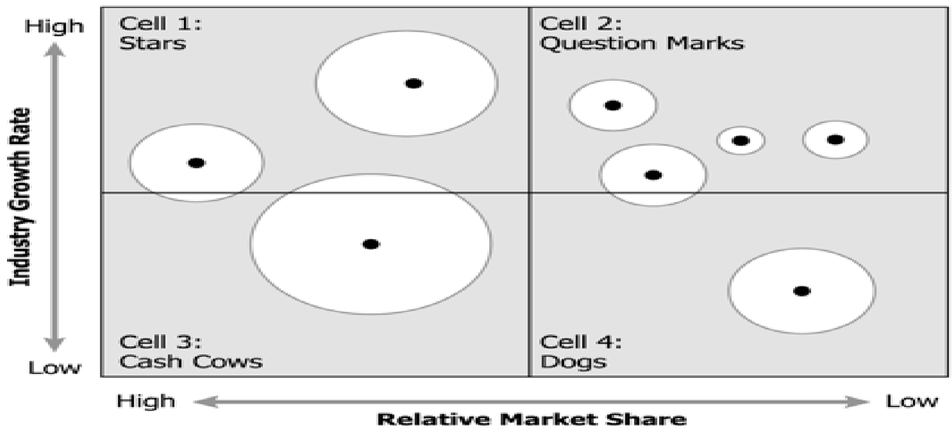
1. Company's relative market share of the business, representing competitive position of the firm, and
2. The overall growth rate of the business.

The BCG model proposes that for each business activity within the corporate portfolio a separate strategy must be developed depending on its location in a two-by- two matrix of high and low segments on each of the above mentioned axes. These parameters are discussed in detail below.

Relative market share is based on the assumption that the relative competitive position of the company would determine the rate at which the business generates cash. An organisation with a higher relative share of the market compared to its competitors will have higher profit margins and therefore higher cash flows. (This point of view can be debated upon since a high market share, per se, may or may not be linked to high profitability or growth in future). Relative Market Share is defined as the market share of the relevant business divided by the market share of its largest competitor. Thus, if Company X has 10 per cent, Company Y has 20 per cent, and Company Z has 60 per cent share of the market, then X's Relative Market Share is 1/6, Y's Relative market Share is 1/3, and Z's Relative Market Share is 60/20 = 3. Company Z has Company Y as its leading competitor, whereas Companies X and Y have Company Z as their leading competitor.

The selection of the Rate of Growth of the associated industry is based on the understanding that an industrial segment with high growth rate would facilitate expansion of the operations of the participating company. It will also be relatively easier for the company to increase its market share, and have profitable investment opportunities. A high growth rate business provides opportunities to plough back earned cash into the business and further enhance the return on investment. The fast growing business, however, demands more cash to finance its growth. If an industrial sector is not growing, it would be more difficult for the participating company to have profitable investments in that sector. In a slow growth business, increase in the market share of a company would generally come from corresponding reduction in the competitor's market share.

The BCG matrix classifies the business activities along the vertical axis according to the `Business Growth Rate' (mean growth of the market for the product), and the `Relative Market Share' along the horizontal axis. The two axes are divided into Low and High sectors; so that the BCG matrix is divided into four quadrants (refer to Figure 6.1). Businesses falling into each of these quadrants are classified with broadly different strategic categories, as explained below:



Source: Hill and Jones (1998, p. 316)

Cash Cows: The business with low growth rate and high market share are classified in this quadrant. High market share leads to higher generation of cash and profits. The low rate of growth of the business implies that the cash demand for the business would be low. Thus Cash Cows normally generate large cash surpluses. Cows can be `milked' for cash to help to provide cash required for running other diverse operations of the company. Cash Cows provide the financial base for the company. These businesses have superior market position and invariably low costs. But, in terms of their future potential, one must keep in mind that these are mature businesses with low growth rate.

Dogs: If the business growth rate is low and the company's relative market share is also low, the product is classified as Dog. The low market share normally also means poor profits. As the growth rate is also low, attempts to increase market share would demand prohibitive investments. Thus, the cash required maintaining a competitive position often exceeds the cash generated, and there is a net negative cash flow. Under such circumstances, the strategic solution is to either liquidate, or, if possible, harvest or divest the dog business.

Question Marks: Like dogs, Question Marks are products with low market share, but the product has a high growth rate. Because of their high growth, the cash requirement is high, but due to their low market share, the cash generated is also low. As the business growth rate is high, one strategic option is to invest more to gain market share, pushing from low share to high. The question mark business then moves to a Star (discussed later) quadrant, and subsequently has the potential to become a cash cow, when the business growth rate reduces to a lower level. Another strategic option when the company cannot improve its low competitive position (represented by low market share) is to divest the question mark business. These products are called Question Marks because they raise the question as to whether more money should be invested in them to improve their relative market share and profitability, or they should be divested and dropped from the portfolio.

Stars: Products, which have high growth rate and high market share, are called Stars. Such businesses generate as well as use large amounts of cash. The stars generate high profits and represent the best investment opportunities for growth. The best strategy regarding stars is to make the necessary investments and consolidate the company's high relative competitive position.

BCG Matrix-Building Procedure

The Boston Consulting Group suggests the following step-by-step procedure to develop the business portfolio matrix and identify the appropriate strategies for different products.

- Classify various activities of the company into different business segments or Strategic Business Units (SBUs).
- For each business segment, determine the growth rate of the market. This is later plotted on a linear scale.
- Compile the assets employed for each business segment and determine the relative size of the business within the company.
- Estimate the relative market shares for the different business segments. This is generally plotted on a logarithmic scale.
- Plot the position of each business on a matrix of business growth rate and relative market share. A bubble represents the size of the business; a circle with a diameter corresponding to say the assets employed in that business.

For precise plotting, it has been recommended that the radius of a bubble corresponding to a business/product may be defined as:

r = square root of (P * R2)

Where, R = radius of the large circle representing total company sales, and

P = sales of a product as percentage (expressed in decimal) of the total sales.

Arbitrary lines divide the four quadrants. In most of the cases 10 per cent volume growth is the typical dividing line between high and low growth businesses, and a relative market share of 1.5 X may separate Stars from the Question Marks in high- growth industries. On the other hand, the recommended relative market share dividing Cows and Dogs is IX for low growth industries. It is, however, added that these dividing lines are merely approximate guidelines and may be changed if desired.

BCG Matrix-Strategic Implications

Most companies will have different segments scattered across the four quadrants of the BCG matrix, corresponding to Cash Cow, Dog, Question mark and Star businesses.

The general strategy of a company with diverse portfolio is:

I. To maintain its competitive position in the Cash Cows, but avoid over-investing.

II. The surplus cash generated by Cash Cows should be invested first in the Star businesses, if they are not self-sufficient, to maintain their relative competitive position.

III. Any surplus cash left with the company may be used for selected Question Mark businesses to gain market share for them.

IV. Those businesses with low market share, and which cannot be adequately funded may be considered for divestment.

V. The Dogs are generally considered as the weak segments of the company with limited or no new investments allocated to them.

The BCG Growth-share matrix links the industry growth characteristic with the company's competitive strength (market share), and develops a visual display of the company's market involvement, thereby indirectly indicating current resource deployment. (The sale to asset ratio is generally stable over time across industries). The underlying logic is that investment is required for growth while maintaining or building market share. But, while doing so, a strong competitive business in an industry with low growth rate will provide surplus cash for development elsewhere in the Corporation. Thus, growth uses cash, whereas market competitive strength is a potential source of cash.

In a sense, the BCG matrix can be regarded as a pictorial representation of the sources and uses of funds statement. Market Share is considered valuable because it is a source of profits. Profits are the fruit of accumulated experience, giving rise to a cost advantage. The model assumes that high market growth of star businesses wills subsequently slowdown, permitting the market leader to take cash out of the Cow business. Some of the underlying assumptions may not always hold true for some products. For instance, electronic appliances and the so-called fashion goods have very short life cycles, whereas staples like bread have very extended life cycles. These businesses may, therefore, not follow the typical behaviour pattern assumed by the BCG growth-share matrix.

Conditions for Having a Balanced Portfolio

An organisation would want to have a balanced portfolio**:**

- cash cows of sufficient size and/or number that can support other products in the portfolio
- stars of sufficient size and/or number which will provide sufficient cash generation when the current cash cows can no longer do so
- problem children that have reasonable prospects of becoming future stars
- no dogs or, if there are any, there would need to be good reasons for retaining them.

Limitations of the BCG Matrix

The Growth-share BCG Matrix has certain limitations and weak points that must be kept in mind while using portfolio analysis for developing strategic alternatives. These are as discussed below:

1. Predicting Profitability from Growth and Market Share

BCG analysis assumes that profits depend on growth and market share. The attractiveness of an industry may be different from its simple growth rate, and the firm's competitive position may not be reflected in its market share. Some other sophisticated approaches have been evolved to overcome such limitations.

There have been specific research studies, which illustrate that the well-managed Dog businesses can also become good cash generators. These organisations relying on high-quality goods, with medium pricing and judicious expenditure on R&D and marketing, can still provide impressive return on investment of above 20 per cent.

2. Problems in Determining Market Share

There is a heavy dependence on the market share of a business as an indicator of its competitive strength. The calculation of market share is strongly influenced by the way the business activity and the total market are defined. For instance, the market for helicopters may encompass all types of helicopters, or only heavy helicopters or only heavy military helicopters. Furthermore, from a geographical point of view the market may be defined on a worldwide, national or even regional basis. In case of complex and interdependent industries, it may also be quite difficult to determine the market share based on the sales turnover of the final product only.

3. Effect of Experience Ignored

In the BCG approach, businesses in each of the different quadrants are viewed independently for strategic purposes. Thus, Dogs are to be liquidated or divested. But, within the framework of the overall corporation, useful experiences and skills can be acquired by operating low-profit Dog businesses, which may help in lowering the costs of Star or Cash Cow businesses. This may contribute to higher corporate profits.

4. Disregard for the Human Aspect

The BCG analysis, while considering different businesses does not take into consideration the human aspects of running an organization. Cash generated within a business unit may come to be symbolically associated with the power of the concerned manager. Consequently, the manager running a Cash Cow business may be reluctant to part with the surplus cash generated by his unit. Similarly, the workers of a Dog business which were decided to be divested may react strongly against changes in the ownership. They may deem the divestiture as a threat to their livelihood or security. Thus, BCG analysis could throw up strategic options, which may or may not be easy to implement.

5. Modifications in BCG Approach

It was in 1981 that the Boston Consulting Group realised the limitations of equating market share with the competitive strength of the company. They have admitted that the calculation of market share is strongly influenced by the way business activity and the total market domain is defined. A broadly defined market will give lower market share, whereas a narrow market definition will result in higher- market share resulting in the company as the leader. It was, therefore, recommended that products should be regrouped according to the manufacturing process to highlight the economies of scale manufacturing, instead of stressing the market leadership.

On the other hand, BCG still maintains that for branded goods it is important to be the market leader so that the advantages of economies of scale and price leadership can be fully utilised. But they also concede that such advantages may still be achieved even if the company is not the largest producer in the industry. Some other versions of portfolio analysis have, however, developed much beyond these minor modifications of BCG analysis.

Activity 1

Draw the BCG matrix for Gujerat Milk Marketing Federation (GCMMF) in the space below. Discuss what you need to do to optimise profits for the next 5 years. Draw up your Marketing plan describing the 4Ps for your Stars and Question Marks. Would you like to maintain the dogs or do you see a future for them. Explain and justify.

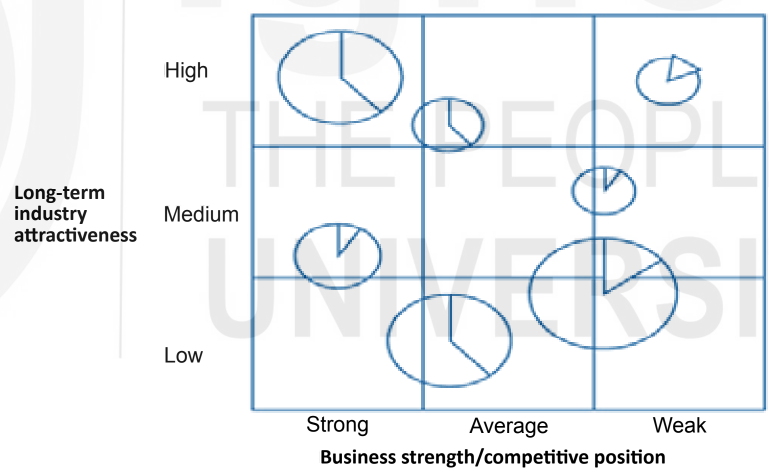
Space for Diagram

4.6 GENERAL ELECTRIC (OR MCKINSEY) MATRIX

This matrix uses market attractiveness as not merely the growth rate of sales of the product, but a compound variable dependent on different factors influencing the future profitability of the business sector. These different factors are either subjectively judged or objectively computed on the basis of certain weights, to arrive at the Market Attractiveness Index. The Index is thus based on a thorough environmental assessment influencing the sectoral profitability.

The Market Attractiveness/SBU Strength Matrix: As can be seen in the diagram below, the matrix has four dimensions on two axes.

- Industry attractiveness:-which includes the size, growth, diversity, profitability and competitive structure of the industry, as well as relevant political, economic, social and technological factors.
- Business or competitive strengths:-another composite dimension including size, growth, share, position, profitability, image, strengths and weaknesses.
- Size of industry:-products or services (strategic business units) are entered onto the policy matrix as circles, with size proportional to turnover size of circle.
- Share of industry:- represented by a segment of the circle.



Each axis should be defined in terms that are meaningful to that company. Some examples are given below.

Competitive Strengths	Industry Attractiveness
• Relative market share • Distribution/brand strengths • Technology strengths • Innovation/management • Profit margins relative to competitors • Ability to compete on price and quality • Knowledge of customer and market • Calibre of management	▪ Market size, growth, diversity ▪ Inflation recovery ▪ Technology protection ▪ Socio/political risks ▪ Economies of scale ▪ Seasonality ▪ Cyclicality ▪ Barriers to entry and exit

Strategic choices

This matrix can also be used to develop and clarify strategic choices:

a. depending on where a unit is positioned in the matrix, its broad strategic mandate will be to invest/build, hold or harvest

b. arrows can be attached to the circles showing the direction in which the strategist wants the product to move

c. the direction of movement can often be changed by management action for example; competitive strength could be increased if resources were directed at technological innovation.

The implied strategies may be as shown in the chart below:

	Business strength/Competitive position		
	Strong	**Average**	**Weak**
Long-term industry attractiveness High	• Enhance leadership • Diversify product/market segment • Re-invest most aggressively in technology, capacity and marketing	• Hold leadership position • Leverage strengths into more attractive segments • Use economies of scale to outspend competition	• Maintain leadership in more attractive segments • Harvest work segments • Shrink production line
Long-term industry attractiveness Medium	• Avoid 'me too' strategy • Segment market carefully • Differentiate product, service or business approach • Concentrate on re-investment	• Increase level segmentation to differentiate strategies between growth and harvest • Avoid perpetrating high shared overheads	• Harvest by pricing up, selling shares and cutting costs aggressively through line pruning
Long-term industry attractiveness Low	• Apply high degree of segment focus • Seek advantage, e.g. proprietary technology	• Harvest by sale to larger company in market	• Divest or liquidate

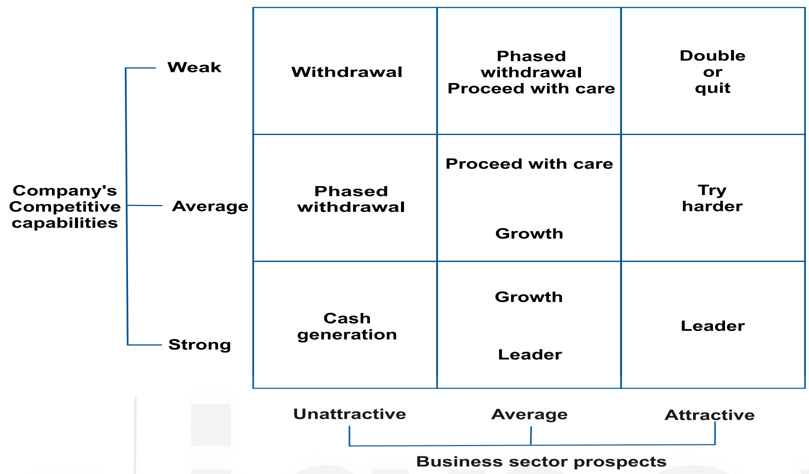
Strengths and Weaknesses of SBU Strength Matrix

There are strengths and weaknesses of this approach.

- The use of multiple criteria provides sounder judgement.
- The criteria used can be tailored to business level or business type and the matrix is relevant to many business situations.
- However, the analysis is not totally quantitative and judgements need to be made.
- To be effective the matrix needs sophisticated users.

4.7 SHELL'S DIRECTIONAL POLICY MATRIX

The Directional Policy Matrix (DPM) initially worked out by the Shell group is a framework which can be used to classify and categorise an organisation's business activities in terms of its strengths, capabilities or market position, and the way it perceives markets to be attractive. The basic structure of a DPM is illustrated in the diagram below:



The purpose of the matrix is to diagnose an organisation's strategic options in relation to those two composite dimensions; business strengths and market attractiveness. The DPM, therefore, enables organisations to conduct an analysis of their portfolio of products or areas of operation. The analysis is performed, first, according to the potential each product area or business has to achieve the organisation's objectives and second, according to the organisation's ability to take advantage of the range of opportunities it faces. The matrix requires its users to identify a number of factors which will act as indicators of the attractiveness of a market or opportunity to them and, similarly, a number of factors which will act as indicators of organisational strengths. These factors will obviously vary from one product or business area to another, since what will be attractive to one activity may not be attractive in another area of operation, or, at least, may not be equally attractive. Similarly, a strength or capability in one area may not exist or be strength in another.

In comparison with other portfolio analysis tools, such as the Boston Matrix, the DPM is a more sophisticated mechanism. Like its forerunner, the Boston Matrix, it can be used to derive quantitative comparisons between areas of activities, but, in addition, is able to take into account a much wider range of decisional influences. In fact, the DPM was originally conceived by General Electric and developed by McKinsey (management consultants), and later Shell, as a means of overcoming some of the limitations the Boston Matrix was perceived to have. It is usually drawn as a 3 x 3 box matrix, rather than the more standard 2 x 2 format, in order to encompass the range of strategic options it covers. In the end, the number of lines drawn is irrelevant. What is more important is defining the substance of the matrix and its axes, and adopting an orderly methodology for its application

Strategic Business Unit (SBU):

Since the purpose of performing a DPM analysis is to provide a basis for determining policy and allocating resources for the alternative products or business within an organisation, it is important to consider the organisational level at which the analysis should be conducted. This is normally taken as being that of the 'strategic business unit' (SBU).

The most common definition of a SBU is that it will:

- Have common segments and competitors for most of its products.
- Operate in external markets.
- Be identifiable as a discrete and separate unit.
- Be managed by people who will have control over most of the areas critical to success

Thus, the process of defining an SBU can be applied all the way down to product or departmental level. It is, therefore, possible to use the DPM for any unit that has within it a number of different variables that can be usefully plotted using a two-dimensional matrix. There must obviously be two or more markets or segments between which managers wish to choose, and these can be either existing or potential markets. It is usually felt that there should be at least three and a maximum of ten areas for analysis if using the DPM is to be of value.

Rather than simply include market growth and share, the DPM considers a range of factors including the following.

Business Sector Prospects

- *Market*: demographic factors, growth, seasonality, maturity.
- *Competition*: number and size of competitors, price competition, barriers to entry, substitutes.
- *Technology*: sophistication, rate of change, lead time, patents.
- *Economic*: leverage, capital intensity, margins.
- *Government*: subsidies/grants, purchases, protection, regulation, taxation.
- *Geography*: location, markets, communications, environment.
- *Social*: pressure groups, trade unions, availability of labour.

Competitive Capabilities

- *Market*: share, growth, product maturity, product quality, product mix, marketing ability, price strategy, customer loyalty.
- *Technological*: skills, patent protection, R&D, manufacturing technology.
- *Production*: costs, capacity utilisation, inventory control, maintenance, extent of vertical integration.
- *Personnel*: employee quality, top management quality, industrial relations, trade union strength, training, labour costs.
- *Financial*: resources, capital structure, margins, tax position, financial control, investment intensity

Possible DPM Strategic Choices

The cell labels shown in the diagram above represent possible strategic choices or types of resource deployments most appropriate for the firm, given its score on each of the two axes. More specifically these cell labels have the following implications.

- ***Withdrawal***: - probably already losing money; net cash flow negative over time. Losses may be minimised by divesting or even liquidation.

- ***Phased withdrawal***: - probably not generating sufficient cash to justify continuation; assets can be redeployed.
- ***Cash generation:*** - equivalent to a **'cash cow'** in the BCG grid. This cell would be occupied by a firm or product in later stages of the life cycle that does not warrant heavy investment, but can be 'milked' of cash due to its strong competitive position.

- ***Proceed with care:*** - similar to a **'problem child'**; firms falling in this sector may require some investment support but heavy investment would be extremely risky.
- ***Growth:*** - a firm, product or SBU in these sectors would call for investment support to allow growth with the market. It should generate sufficient cash on its own.
- ***Double or quit:*** - units in this sector should become 'high fliers' in the not-too-distant future. Consequently those in the upper right corner of this cell should be singled out for full support. Others should be abandoned.
- ***Try harder:*** - external financing may be justified to push a unit in this sector to a leadership position. However, such a move will require judicious application of funds.
- ***Leader: -*** the strategy for this segment is to protect this position by external investment (funds beyond those generated by the unit itself - occasionally); earnings should be quite strong and a major focus may be maintaining sufficient capacity to capitalise on strong demand.

Strategic Movements

In addition to considering the position on the **directional policy matrix** in static terms, changes over time need also to be considered.

- Ideally, products in the cash-generating sectors should be able to finance expenditure on products in the attractive business/weak position sectors, so as to move them to the attractive business/strong position sectors.
- Later these products move down to become cash generators themselves and the cycle is completed.

4.8 PROFIT IMPACT OF MARKETING STRATEGY (PIMS MODEL)

A program for the Profit Impact of Market Strategy (PIMS) was started at General Electric, and was later used by the Strategic Planning Institute. The PIMS program analyses data provided by member companies to discover `general laws, which determine the business strategy in different competitive environments producing different profit results'. Unlike the earlier approaches using judgement for multi dimensional factors, the SPI uses multidimensional cross-sectional regression studies of the profitability, of more than 2,000 businesses. It then develops an industry characteristic, business average profitability, and compares it with the performance in the concerned company.

This model uses statistical relationship estimated from past experience, in place of the judgmental weights assigned for the importance of different factors behind Market Attractiveness and Competitive Position in previous approaches. This scientific objective approach has been criticised saying that the analysis of relationships is based on a heterogeneous population, i.e., different types of business, taken at different time periods. Profitability is closely linked with market share. A 10% improvement in profitability is linked with 5% improvement in Return on Investment (ROI). This has since been rationalised by a number of arguments, such as `the Experience Curve Effect', which implies reduction in average cost with increase in accumulated production. The larger company can use better quality management, and thus can exercise greater market power.

4.9 ARTHUR D. LITTLE COMPANY'S MATRIX

Arthur D. Little Company's matrix links the stages of the product life cycle with the business strength. On the vertical axis, the businesses are classified with respect to their business strength: Weak, Tenable, Favourable, Strong, or Dominant: Along the horizontal axis four stages in the life cycle, Embryonic, Growth, Mature and Decline are marked.

Arthur D Little Company's Matrix

		Industry Life Cycle Stage			
		Embryonic	Growth	Mature	Aging
Competitive Position	Dominant	All out push for share. Hold Position	Hold Position. Hold Share.	Hold Position. Grow with industry	Hold Position.
	Strong	Attempt to improve position. All out push for share	Attempt to improve position. Push for share.	Hold Position. Grow with industry	Hold Position or Harvest.
	Favorable	Selective or all out put for share. Selective attempt to improve position.	Attempt to improve position. Selective push for share.	Custodial or maintenance. Find niche and attempt to protect it.	Harvest, or phased out withdrawal.
	Tenable	Selectively push for position	Find niche and protect it.	Find niche and hang on, or phased out Withdrawal.	Phased out withdrawal, or Abandon
	Weak	Up or out	Turnaround or abandon	Turnaround, ophaned out withdrawal.	Abandon

In the Embryonic and Growth stages the businesses are recommended for Build strategy, except when the business strength is weak. For Mature stage businesses with dominant to favourable strength, HOLD Strategy is recommended. Harvest strategy is proposed for businesses in Decline stage, with Strong or Dominant position. For weaker businesses in Mature/ Decline stage unacceptable ROI is marked.

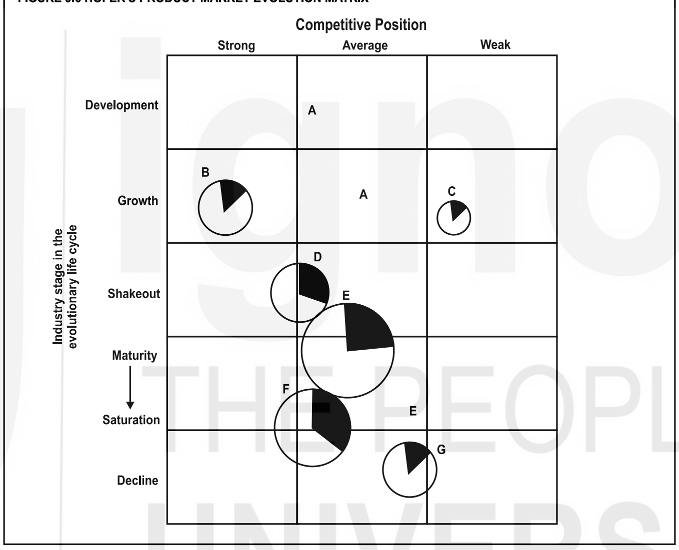
The Advantages and Disadvantages of the ADL matrix

Unlike other models of product portfolio analysis, the ADL matrix is based on an enhanced applicability because it fits to all situations of competition encountered in a marketplace. Also the ADL matrix can be applied to the fragmented industries, holding a small competitive advantage but with a large number of ways of obtaining it (provides multiple ways of differentiation). As such, we can say that the ADL matrix has a high degree of adaptability to situations of a qualitative nature. A first disadvantage is that the matrix does not take into account a number of phenomena that can generate long-term involution in the products life cycle of a company. Another weakness is related to the high level of difficulty in terms of objective evaluation of the ADL model variables. This is often the case for the competitive position indicator. In other words, the difficulty lies in the fact that some factors are qualitative in nature and there is a high risk of bias in their use. In conclusion, we can say that the ADL matrix provides clearer results as a company is more diversified and enables synchronization on decisions relating to competition.

4.10 HOFER'S PRODUCT/MARKET EVOLUTION MATRIX

Charles Hofer has proposed a three-by-three matrix where products are plotted in terms of their product/market evolution and the competitive position. Relative sizes of Industries are shown by circles wherein the market of the company is shaded.

FIGURE 9.6 HOFER'S PRODUCT-MARKET EVOLUTION MATRIX



Accthis model a firm's business is positioned in a 15-cell matrix based on two major variables viz., stage of production-market development and the competitive position. Charles W. Hoffer has suggested a further refinement of GE/ Mckinsey portfolio matrix by identifying companies, particularly new businesses, that are about to accelerate their growth. This matrix is also called 'life-cycle portfolio matrix. An illustrative graph representing Hofer's matrix is given in figure above provides potential strategies for different units placed in the matrix.

Hofer's matrix reflects the stage of development of the product or market. Business units are placed on a grid showing their stage of product-market evolution and their competitive position. Circles represent the industry and the pie wedges represent the market share of the business unit. Hofer's evolution matrix is useful to develop strategies that are appropriate at different stages of the product life cycle. In Hofer's matrix, the vertical axis represents the stages of product-market evolution and horizontal axis represents the SBU's competitive position. In this matrix, three stages of competitive position of SBU (viz., strong, average and week) are shown on horizontal axis. The vertical axis shows the industry's state in the evolutionary life cycle, starting with initial development and passing through the growth, competitive shake-out, maturity, saturation, and decline stages.

1) ***SBU A*** with average competitive position and in development stage holds out prospects for future development deserves expansion and desired financial resources to be allotted to exploit the opportunities.

2) ***SBU B*** with strong competitive position and in growth stage requires to adopt growth strategies to make it a future winner.

3) ***SBU C*** with weak competitive position which is in growth stage of the industry should give lot of attention and requires a careful formulation of marketing strategies to make it more competitive in the industry.

4) ***SBU D*** with moderately strong position is in the shake-out stage can be probable with close attention and careful marketing strategy formulation. This may also requires adoption of growth strategies.

5) ***SBU E*** with average competitive position and in maturity stage of the industry needs to adopt stability strategies.

6) ***SBU F*** with moderately strong competitive position and is in the maturity stage of the industry life cycle, needs the stability, harvest and retrenchment strategies need to be adopted. No further funds to be invested in this SBU. The market strategies require to hold the market position without fall.

7) ***SBU G*** with moderately weak competitive position and is in the decline state of the industry life cycle need to be divested immediately to arrest any cash loss since it is in a position of losing. Revival of this SBU is not suggested. The Hofer's product-market evolution matrix displays business portfolio of an international firm with relative greater degree of accuracy and completeness.

The Hofer's Matrix considered the following Variables:

1. Variable # (a) Market and Consumer Behaviour Variables Like:

i. Buyer needs
ii. Purchase frequency
iii. Buyer concentration
iv. Market segmentation
v. Market size
vi. Elasticity of demand
vii. Buyer loyalty
viii. Seasonality and cyclicality

2. Variable # (b) Industry Structure Variables Like:

i Uniqueness of the product
ii. Rate of technological change in product design
iii. Type of product
iv. Number of equal products
v. Barriers to entry
vi. Degree of product differentiation
vii. Transportation and distribution costs
viii. Price/cost structure
ix. Experience curve
x. Degree of integration
xi. Economy of scale etc.

3. Variable # (c) Competitor Variables Like:

i. Degree of specialization within the industry
ii. Degree of capacity utilization
iii. Degree of seller concentration
iv. Aggressiveness of competition

4. Variable # (d) Supplier Variables Like:

i. Degree of supplier concentration
ii. Major changes in availability of raw materials

5. **Variable # (e) Broader Environment Variables:**
   i. Interest rates
   ii. Money supply
   iii. GNP trend
   iv. Growth of population
   v. Age distribution of population
   vi. Life cycle changes

6. **Variable # (f) Organizations Variables Like:**
   i. Quality of products
   ii. Market share
   iii. Marketing intensity
   iv. Value added
   v. Degree of customer concentration etc.

Hofer developed descriptive propositions for each stage of product life cycle.

For example- in the maturity stage of the product life cycle, Hofer identified the following major determinants of business strategy:

(a) Nature of buyer needs

(b) Degree of product differentiation

(c) Rate of technological change in the process design

(d) Ratio of market segmentation

(e) Ratio of distribution costs to manufacturing

(f) Value added

(g) Frequency with which the product is purchased

Hofer, thereafter formulated normative contingency hypothesis using the above major determinants.

An example for the maturity stage is when:

(a) Degree of product differentiation is low.

(b) The rate of buyer needs is primarily economic.

(c) Rate of technological change in process design is high.

(d) Purchase frequency is high.

(e) Buyer concentration is high.

(f) Degree of capacity utilization is low.

Then the business firms should:

1. Allocate most of their R&D funds to improvements in process design rather than to new product development.
2. Allocate most of their plant and equipment expenditures to new equipment purchases.
3. Seek to integrate forward or backward in order to increase the value they added to the product.
4. Attempt to improve their production scheduling and inventory control procedures in order to increase their capacity utilization.
5. Attempt to segment the market.
6. Attempt to reduce their raw material unit costs by standardizing their product design and using interchangeable components throughout their product line in order to qualify for volume discount.

4.11 UTILITY OF DISPLAY MATRICES

It is important to note that whereas the specific names of axes differ from matrix to matrix, they are based on quite similar principles. In one form or another, most portfolio approaches try to correlate industry growth or profitability with market share, either as a direct single variable or as an index based on multiple variables. Further, these matrices are meant to facilitate a graphic display of the diversity of an organization rather than to provide a precise analytical tool. The matrices help to raise critical questions about improper deployment of funds and gross mismatches in businesses, and not so much to give precise answers as to where and how the next unit of money should be used.

Experience shows that portfolio analysis is not applicable where market share is not so critical, or the capital cannot be easily withdrawn. Similarly extra care is required in utilising portfolio analysis if value added is low or cost can be lowered without experience, or technology is transferred rapidly by suppliers. Seasonality of and cyclic businesses, IPR restrictions and `low economies of scale also complicate the strategic outcomes from portfolio analysis.

To conclude, the models discussed here must be used to stimulate managers to think about their businesses in an integrated manner.

Activity 2

Visit a local branch office of any diversified group company and gather information about its product portfolio .Give below your analysis on how well the company has organized its portfolio to serve market needs.

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4.12 SUMMARY

Portfolio Analysis is a very important task for a Product Manager. It provides a framework for analysing the mutual compatibility of diverse operations of an organisation. The portfolios of operations need to be balanced with respect to net cash flows, state of development, and the risks associated with each business activity. After discussing the need for balancing the portfolio with respect to these aspects, different types of display matrices have been introduced in this unit.

The Boston Consulting Group's Growth-Share Matrix, being the pioneering model, was first taken up for detailed discussion. A methodology for building up the BCG Matrix was proposed. The strategic implications, balancing of portfolios, and variations with time were covered next. Some of the limitations of the BCG Matrix with respect to determination of profitability, market share and lack of consideration for the experience curve synergy and human aspects associated with strategic actions were discussed along with some of the modifications proposed by the Boston Consulting Group.

Features of other display matrices, such as General Electric's Strategic Business Planning Grid, Shell's Directional Policy Matrix, Strategic Planning Institute's Matrix (PIMS Model) and matrices based on product life-cycle or market evolution were explained in the context of their departure from the BCG Matrix. The overall utility of the Product Portfolio was also explained.

4.13 SELF- ASSESSMENT QUESTIONS

1. What basic considerations have to be kept in mind while balancing portfolios?
2. Explain the methodology of constructing the BCG Matrix.
3. Discuss the limitations of the BCG Matrix.
4. How does the GE Planning Grid differ from the BCG Matrix?
5. Explain Shell's Directional Policy Matrix.
6. Explain the uses of: a) PIMS Model b) Arthur D. Little Company's Matrix c) Hofer's Product Market Evolution Matrix.

4.14 REFERENCES/FURTHER READINGS

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